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DOCTORAL THESIS

Data Representation and Learning Strategies for Inferring Gene Regulatory Networks from Time-Series Gene Expression Data

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Declaration of Authorship

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Abstract

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Doctor of Philosophy

Data Representation and Learning Strategies for Inferring Gene Regulatory Networks from Time-Series Gene Expression Data

by Cao TUAN ANH

Understanding the complex mechanisms governing gene regulation is a central challenge in systems biology. Gene Regulatory Networks (GRNs) provide a powerful framework to describe how transcription factors and other regulatory elements coordinate gene expression dynamics across cellular processes. However, accurately reconstructing GRNs from gene expression data remains difficult due to the intrinsic noise, limited sample size, and nonlinear dependencies inherent in biological systems. This dissertation aims to enhance the inference of GRNs from time-series gene expression profiles through three complementary methodological contributions.

Chapter 1 introduces the research background and establishes the foundation of this dissertation. It presents the problem statement, discusses the major research challenges in GRN inference from time-series gene expression data, and provides an overview of existing and proposed methodological approaches. The chapter also outlines the organization of the dissertation, establishing the relationships between the identified research challenges and the methodological contributions developed in subsequent chapters.

In Chapter 2, a novel mutual information-based multi-level discretization (MIDNI) approach is proposed to infer GRNs from temporal gene expression data. Unlike conventional discretization or correlation-based methods, MIDNI adaptively partitions gene expression ranges into multiple information-preserving levels. This enables the capture of nonlinear and time-dependent regulatory relationships, improving sensitivity and robustness in network reconstruction.

Chapter 3 introduces a cross-size network reconstruction model designed to improve generalization across datasets of varying scales and dimensions. The proposed framework leverages a scalable parameter sharing mechanism and regularized structure optimization to enable the inference of GRNs that remain consistent and biologically meaningful even when trained on datasets of different sample sizes. This chapter demonstrates how structural priors and cross-size alignment can mitigate overfitting and enhance model transferability.

In Chapter 4, a synthetic time-series data generation framework based on autoencoders is developed to address the scarcity and imbalance of gene expression data. By learning latent representations that preserve both temporal and regulatory dependencies, the model synthesizes realistic gene expression trajectories that can be integrated with real datasets to strengthen GRN inference. Experimental results show that augmenting limited datasets with generated time-series samples significantly enhances network reconstruction accuracy and stability.

Finally, Chapter 5 provides a general discussion of the dissertation's methodological contributions and experimental findings. It examines the implications of the proposed approaches for GRN inference, discusses their limitations, and identifies future research directions. This chapter also synthesizes the relationships among the three contributions and considers opportunities to further improve network inference through more effective data representation, and data generation strategies.

Together, the five chapters form a coherent investigation into advancing GRN inference from time-series gene expression data. The proposed methodologies address key limitations in discretization and data availability, providing complementary strategies for reconstructing gene regulatory relationships.

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Chapter 1

General Introduction

1.1 Background and Motivation

Understanding how genes interact to control cellular processes is one of the central goals of systems biology. Genes rarely function in isolation; instead, they form intricate regulatory systems in which transcription factors, signaling molecules, and noncoding RNAs coordinate to regulate expression levels dynamically. These complex interactions are often represented as Gene Regulatory Networks (GRNs), where nodes correspond to genes and edges represent regulatory relationships such as activation or repression. Accurately reconstructing GRNs provides fundamental insights into the mechanisms underlying development, disease progression, and cellular responses to environmental changes.

Over the past decades, the rapid advancement of high throughput sequencing technologies such as microarrays and RNA-seq has made it possible to measure the expression levels of thousands of genes simultaneously across multiple time points. These time-series gene expression datasets capture the dynamic behavior of transcriptional regulation, offering a valuable resource for inferring causal and temporal dependencies among genes. However, despite these advances, the task of accurately inferring GRNs from time series data remains a formidable challenge due to noise, limited sample sizes, and the nonlinear and stochastic nature of gene regulation.

The motivation for this research arises from the need to develop robust, scalable, and data efficient approaches to infer biologically meaningful networks from complex and often sparse time series data. A deeper understanding of regulatory relationships can significantly enhance predictive modeling in systems biology and contribute to practical applications such as biomarker discovery, drug target identification, and personalized medicine.

1.2 Problem Statement

Inferring GRNs from time-series gene expression data involves uncovering the structure of gene–gene regulatory relationships based on observed expression dynamics. However, several major obstacles hinder this process:

- Nonlinearity and temporal dependency: Gene regulation often involves nonlinear relationships and delayed interactions that simple correlation-based or linear models fail to capture.

- Discretization and data representation: Many information-theoretic methods require discretized expression values, yet choosing an appropriate discretization strategy that preserves temporal information and minimizes information loss remains difficult.
- Scalability and dataset heterogeneity: Biological datasets vary in scale, dimension, and noise characteristics, making it hard to construct models that generalize across studies of different sample sizes.
- Data scarcity and imbalance: In many biological experiments, the number of time points or replicates is limited, which restricts the model's ability to learn robust patterns and increases overfitting risk.

These challenges motivate the development of a comprehensive framework that can (i) effectively model nonlinear and dynamic dependencies, (ii) maintain performance across datasets of varying scales, and (iii) enhance learning through data synthesis or augmentation.

1.3 Research Objectives and Scope

The overall objective of this dissertation is to advance the inference of gene regulatory networks from time-series gene expression profiles by addressing the limitations of discretization, model scalability, and data scarcity. Specifically, the research focuses on three main directions:

- Developing a mutual information–based multiple level discretization framework to improve sensitivity and robustness in network inference from temporal gene expression data.
- Designing a cross-size scalable model that allows consistent and transferable network reconstruction across datasets of different dimensions and sample sizes.
- Proposing an autoencoder-based synthesis approach for generating realistic time-series gene expression data to augment limited datasets and enhance network inference performance.

The research is primarily computational and data-driven, combining techniques from information theory, machine learning, and systems biology. While the study is validated through benchmark datasets and simulation frameworks, the ultimate goal is to build methodological foundations that can be extended to real-world biological systems.

1.4 Research Challenges

The task of GRN inference from time-series data poses unique challenges that require methodological innovations:

- Capturing nonlinear dependencies: Biological systems are inherently nonlinear; thus, linear correlation or regression-based models are often insufficient. Mutual information (MI) offers a nonparametric approach to quantify nonlinear associations, but its effectiveness depends strongly on data discretization quality.
- Balancing information loss and noise: Discretization tends to simplify expression signals but risks removing subtle temporal dynamics. An adaptive multiple level discretization scheme can help preserve information content while improving interpretability.
- Scalability across datasets: In many real-world cases, training data and test data differ in size or sampling frequency. Models that rely heavily on data scale often fail to generalize. Therefore, a cross-size learning framework is essential for robust reconstruction across heterogeneous datasets.
- Limited and imbalanced data: Biological experiments often produce short time-series with few replicates. Data synthesis using generative deep learning such as autoencoders can help overcome this issue by producing realistic synthetic samples that mimic genuine gene expression dynamics.

Addressing these challenges collectively forms the foundation of this dissertation.

1.5 Overview of Methods and Thesis Organization

This dissertation is composed of three main studies, each corresponding to a publication that tackles a specific aspect of the GRN inference problem.

- Chapter 1: ***General Introduction***

  This chapter provides the background and motivation for gene regulatory network inference in systems biology. It reviews fundamental concepts of gene regulation, time-series gene expression analysis, and existing computational approaches for GRN reconstruction. The chapter further formulates the core problem addressed in this dissertation and discusses key research challenges, including data sparsity, noise, nonlinearity, scalability, and the trade-off between structural and dynamic accuracy.

- Chapter 2: ***A Mutual Information-based Multiple Level Discretization Network Inference from Time-series Gene Expression Profiles***

  This chapter introduces a novel discretization framework that integrates adaptive binning with information-theoretic dependency estimation. The method enhances MI-based GRN inference by preserving dynamic signal patterns while reducing noise sensitivity.

- Chapter 3: ***A Representaion Learning Framework for Gene Regulatory Network Inference from Time-series Expression Data***

The second study presents a scalable modeling strategy that aligns GRN structures inferred from datasets of varying sizes. By applying regularized structure optimization and parameter sharing, the model achieves improved consistency and generalization across datasets.

- Chapter 4: ***Synthesizing Time-series Gene Expression Data to Enhance Network Inference Performance Using the Autoencoder***

  The third study focuses on data augmentation through deep learning. A temporal autoencoder is designed to learn compact latent representations and generate synthetic gene expression trajectories. These synthetic data are combined with real samples to improve inference accuracy and stability.

Finally, Chapter 5 concludes the dissertation with an integrated discussion of findings, contributions, and potential directions for future research in gene regulatory network modeling and computational biology.

1.6 Summary

In summary, this dissertation seeks to advance the field of gene regulatory network inference by proposing a unified and systematic framework that integrates discretization, scalability, and data synthesis. Through these three interconnected studies, the research aims to contribute novel computational methodologies for decoding the dynamic and nonlinear mechanisms of gene regulation from time-series gene expression data, ultimately enhancing our ability to model and interpret complex biological systems.

Chapter 2

A Mutual Information-based Multiple Level Discretization Network Inference from Time-series Gene Expression Profiles

2.1 Abstract

Inferring GRNs from time-series gene expression data remains a fundamental yet challenging problem in systems biology because it enables researchers to decipher the dynamic transcriptional programs that govern cellular behavior. Information theoretic approaches, particularly those based on mutual information (MI), have been widely used due to their ability to capture nonlinear and nonparametric dependencies among genes. However, the performance of MI-based inference often suffers from inappropriate discretization of continuous expression profiles, which can introduce information loss, obscure temporal patterns, and reduce the accuracy of reconstructed regulatory interactions.

To overcome these limitations, this study introduces a Mutual Information based on Multi-Level Discretization Network Inference (MIDNI) framework tailored for GRN inference from temporal gene expression data. The framework adaptively partitions expression values into informative levels by jointly considering local statistical variance and global entropy characteristics. This dual criteria strategy preserves essential temporal dynamics while mitigating the influence of stochastic noise, thereby yielding more reliable MI estimates and improving the detection of regulatory relationships.

The proposed method was rigorously evaluated using benchmark datasets generated by GeneNetWeaver (GNW) and employed in the DREAM network inference challenges, which represent widely accepted standards for assessing GRN reconstruction algorithms. Comparative experiments demonstrated that MIDNI consistently outperforms traditional discretization techniques such as equal-width and equal frequency binning. Notably, improvements were observed across key evaluation metrics, including precision and recall.

Overall, the results indicate that MIDNI provides a principled and robust discretization strategy for MI-based GRN inference from time-series gene expression data. Its resilience to noise, sampling variability, and limited sample sizes makes

it particularly suitable for realistic biological datasets. By enhancing the quality of MI estimation and preserving critical temporal information, the proposed framework substantially advances the accuracy and interpretability of dynamic GRN reconstruction.

2.2 Introduction

GRNs represent the intricate set of interactions among genes that collectively govern transcriptional activity, thereby orchestrating the functional behavior of organisms. Deciphering the structure and dynamics of GRNs is crucial for understanding both normal cellular processes and the molecular basis of diseases. Despite significant progress in systems biology, accurately reconstructing GRNs remains an enduring challenge due to the complexity, noise, and high dimensionality of genomic data.

Many computational strategies have been developed to address the GRN inference problem, each relying on different modeling assumptions. Data-driven approaches constitute a major class of GRN reconstruction techniques, where gene–gene dependencies are estimated directly from the observed data. Within this class, correlation-based measures are frequently used to quantify associations between pairs of gene expression vectors. Among these, the weighted gene co-expression network analysis (WGCNA) (B. Zhang and S. Horvath, 2005) is one of the most widely adopted correlation-based frameworks and has consistently demonstrated reliable performance. Nonetheless, correlation coefficients are limited in their ability to capture complex statistical dependencies, particularly nonlinear relationships between gene expression profiles. To overcome this constraint, mutual information (MI) has been introduced as a more expressive information-theoretic measure for detecting regulatory associations. The relevance network (Butte and Kohane, 2000) represents one of the earliest MI-based approaches, estimating MI for all gene pairs and inferring regulatory interactions by thresholding these scores. The context likelihood of relatedness (CLR) algorithm (Faith et al., 2007) extends this idea by using empirical MI distributions to perform adaptive background correction, thereby reducing spurious correlations and indirect effects. An algorithm for the reconstruction of gene regulatory networks in a mammalian cellular context (ARACNE) (Zoppoli, Morganella, and Ceccarelli, 2010) is another influential information-theoretic method that identifies direct regulatory dependencies by eliminating interactions that can be explained by indirect relationships among genes. Similarly, MRNET (Meyer, Lafitte, and Bontempi, 2007) employs a maximum relevance/minimum redundancy feature selection strategy to identify regulatory targets using MI-based metrics. Although these approaches have achieved notable success, their primary limitations lie in computational complexity and scalability, making them resource-intensive and less effective when applied to large-scale biological networks.

To address computational burdens and accelerate inference, many studies have transformed continuous gene expression values into Boolean representations and used these discretized datasets as inputs for GRN reconstruction. For example, the Best-Fit algorithm (Lähdesmäki, Shmulevich, and Yli-Harja, 2003)

was introduced to solve the Consistency Problem by identifying one or more Boolean networks that align with observed data. Another Boolean-based approach (Haider and Pal, 2012) infers GRNs from limited time-series transcriptomic or proteomic data by integrating prior biological information, attractor constraints, and robustness considerations, enabling the reconstruction of Boolean networks that explain observed state transitions. The mutual information–based Boolean network inference (MIBNI) method (Barman and Y. K. Kwon, 2017) also employs binarized data to identify initial regulatory gene sets using MI-based feature selection, followed by iterative improvements in dynamic prediction accuracy through gene-pair swapping. An enhanced version of MIBNI, the genetic algorithm–based Boolean network inference (GABNI) approach (Barman and Y. K. Kwon, 2018), expands the search space using a genetic algorithm, improving performance particularly for large-scale networks. However, GABNI remains constrained by the limited representational capacity of the Boolean regulatory function model. To overcome this issue, a hybrid genetic algorithm–neural network model for Boolean network inference was proposed (Barman and Y. K. Kwon, 2020), in which a neural network replaces the incomplete Boolean truth table used in GABNI. Despite these advancements, Boolean network–based methods still suffer from information loss inherent in coarse discretization, which can degrade inference performance in certain conditions.

To mitigate information loss while preserving the computational efficiency of Boolean-based approaches, we introduce a novel method termed mutual information based on multiple level discretization network inference (MIDNI). This framework discretizes gene expression values using either two or three levels, selected according to the intrinsic distribution of each gene. Restricting the discretization to a maximum of three levels avoids excessive computational costs while ensuring that genes with more complex expression patterns retain essential information. The validity index (MacQueen, 1967; Ahn et al., 2019; Ray and Turi, 1999; Halkidi, Batistakis, and Vazirgiannis, 2001) is used to determine, for each gene, whether binary or ternary discretization is more appropriate, enabling an adaptive and data-driven representation strategy.

Following discretization, gene expression profiles are processed using the mutual information–based feature selection (MIFS) approach (Barman and Y. K. Kwon, 2017), which approximates the multivariate MI between a target gene and its candidate regulators. We also incorporate a SWAP subroutine, a greedy procedure that iteratively exchanges genes between the selected and unselected sets to refine the regulatory candidate group. This iterative MIFS–SWAP framework was originally introduced in our earlier Boolean inference method (MIBNI) (Barman and Y. K. Kwon, 2017), and in this work, we extend it to accommodate multi-level discretized data.

We evaluate the performance of MIDNI using both synthetic and real gene expression datasets and compare it against several well-known GRN inference algorithms, including MICRAT, DBN, and MIBNI. Experimental results indicate that MIDNI consistently achieves superior dynamic accuracy, in addition to performing effectively in noisy data scenarios. These findings demonstrate that MIDNI is a robust and reliable framework for GRN inference from gene expression data.

2.3 Materials and Methodology

2.3.1 Materials

A discretization network model

In this study, we utilize a discretization-based network model to characterize gene regulatory networks. A discretized network is formulated as a directed graph $G(V, A)$, in which $V = \{v_1, v_2, \ldots, v_n\}$ denotes the set of genes and, $A = \{(v_i, v_j)\} \subseteq V \times V$ represents regulatory interactions. The state of a gene v at time t, is written as $V(t)$, can take one among l discrete values $\{0, 1, 2, \ldots, l-1\}$. When all genes share $l = 2$, the model reduces to a Boolean network.

Consider a target gene $v \in V$ regulated by k genes, $u_1, u_2, \ldots, u_k$ ($\forall u_i \in V$). Let E be the set of discretized expression values of v and E_i the corresponding set for each regulator u_i. The next state of v, denoted as $v(t+1)$ is determined by a discrete function

$$f : E_1 \times E_2 \times \cdots \times E_k \to E$$

that maps the expression levels of the k regulators at the time t to the new state of v. Thus, the update rule for v is expressed as:

$$v(t+1) = f(u_1(t), u_2(t), \ldots, u_k(t)).$$

In this work, we assume a one-step time lag. The number of all possible functions corresponding to f is

$$l^{\prod_{i=1}^{k} l_i},$$

where l and l_i are the cardinalities of E and E_i, respectively.

The discretization network inference problem

The discretization network inference task focuses on identifying both the interaction set and the update functions using time-series gene expression data. The quality of an inferred model is judged by comparing the predicted expression trajectory with the observed measurements. Let $v'(t)$ denote the predicted state of gene v at time t produced by the inferred discretization network.

The gene-wise dynamics consistency $C(v, v')$ represents how closely the predicted sequence $v'(t)$ matches the observed sequence $v(t)$. It is computed using the following expression:

$$C(v, v') = \frac{\sum_{t=p}^{T} I(v(t) = v'(t))}{T - p},$$

where T refers to the total number of time steps, p is the update lag ($p = 1$), and $I(\cdot)$ is an indicator function that returns 1 when the given condition is met, and 0 otherwise. Since the model assumes a one-step delay, the comparison begins at $t = 2$.

The overall dynamics accuracy of the inferred regulatory network is obtained by averaging the gene-wise consistency scores across all genes:

$$\text{Dynamic Accuracy} = \frac{\sum_{i=1}^{N} C(v_i, v_i')}{N},$$

where N denotes the total number of genes.

Structure performance metrics

When the ground-truth regulatory network is available, the quality of an inferred model can also be assessed in terms of its structural correctness. For this purpose, we evaluate the reconstruction using three standard metrics: precision, recall, and structural accuracy.

Precision quantifies the proportion of correctly identified regulatory links among all predicted links:

$$\text{Precision} = \frac{TP}{TP + FP},$$

where TP represents the number of correctly inferred edges (true positives), and FP corresponds to the number of incorrectly inferred edges (false positives).

Recall measures how many of the true regulatory interactions are successfully recovered by the inference method:

$$\text{Recall} = \frac{TP}{TP + FN},$$

where FN denotes the number of actual edges that were not predicted (false negatives).

tructural accuracy reflects the overall correctness of the inferred structure by considering both positive and negative predictions:

$$\text{Structural Accuracy} = \frac{TP + TN}{TP + FP + FN + TN},$$

where TN is the number of correctly predicted non-interactions (true negatives).

Mutual information

Our approach identifies a set of regulatory variables for each target gene by leveraging fundamental principles from information theory.

To begin, the entropy $H(X)$ of a discrete random variable X is used to quantify the degree of uncertainty associated with its outcomes, and is formally defined as:

$$H(X) = -\sum_{x \in X} p(x) \log p(x),$$

where $p(x)$ denotes the probability of observing the outcome x.

Similarly, the joint entropy $H(X, Y)$ characterizes the combined uncertainty of two discrete random variables X and Y that follow a joint probability distribution $p(x, y)$. It is given by:

$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log p(x, y).$$

Based on these definitions, the mutual information between two discrete variables is expressed as:

$$I(X;Y) = H(X) + H(Y) - H(X,Y).$$

A higher mutual information value indicates a stronger statistical dependency between the variables, and therefore provides a more reliable indicator of potential regulatory influence.

2.3.2 Methodology

In this study, we introduce a novel method, referred to as **MIDNI**, for inferring gene regulatory networks from time-series gene expression data using a multiple-level discretization approach. The overall workflow of the method is illustrated in Figure 2.1.

The input to MIDNI is a real-valued time-series gene expression dataset, which is first transformed into a discretized form using the K-means clustering discretization method (MacQueen, 1967). Each gene's expression profile is partitioned into l discrete levels, where the level l is adaptively determined based on the underlying distribution of the gene's expression values. Specifically, when $l = 2$, the discretized values are labeled 0 (low expression) and 1 (high expression), reflecting a simple binary representation. When $l = 3$, the expression values are labeled 0 (low), 1 (normal), and 2 (high), allowing for a more nuanced representation of gene activity that can better capture subtle regulatory effects and intermediate states.

For each target gene v whose expression exhibits non-zero entropy, the **MIFS** subroutine is employed to identify a subset of k genes in V that are the most informative with respect to v. This step leverages mutual information to quantify the dependency between potential regulators and the target gene, ensuring that candidate regulators capture the strongest statistical signals from the data. Subsequently, the **SWAP** subroutine iteratively refines the selected regulatory set by exchanging genes between the candidate set S and the remainder of the network $W \setminus S$, with the objective of maximizing the gene-wise dynamics consistency. Here, S denotes the candidate regulatory genes identified by MIFS, while W represents the complete set of genes in the network.

This selection and refinement process is performed iteratively by gradually increasing the size of k until an optimal regulatory set S is determined for the target gene v. The optimality criterion is reached when the gene-wise dynamics consistency achieves a perfect score of 1, indicating that the inferred regulators fully capture the observed dynamics of the target gene. If this condition is not met, the process continues until k reaches a user-defined upper bound K, which constrains the maximum number of regulators allowed for any single gene, thereby controlling model complexity and preventing overfitting.

By integrating adaptive multi-level discretization with an information-theoretic feature selection and refinement strategy, MIDNI provides a biologically informed framework for reconstructing regulatory interactions that not only preserves essential dynamics but also improves the interpretability and robustness of inferred networks, particularly when dealing with noisy or limited time-series data.

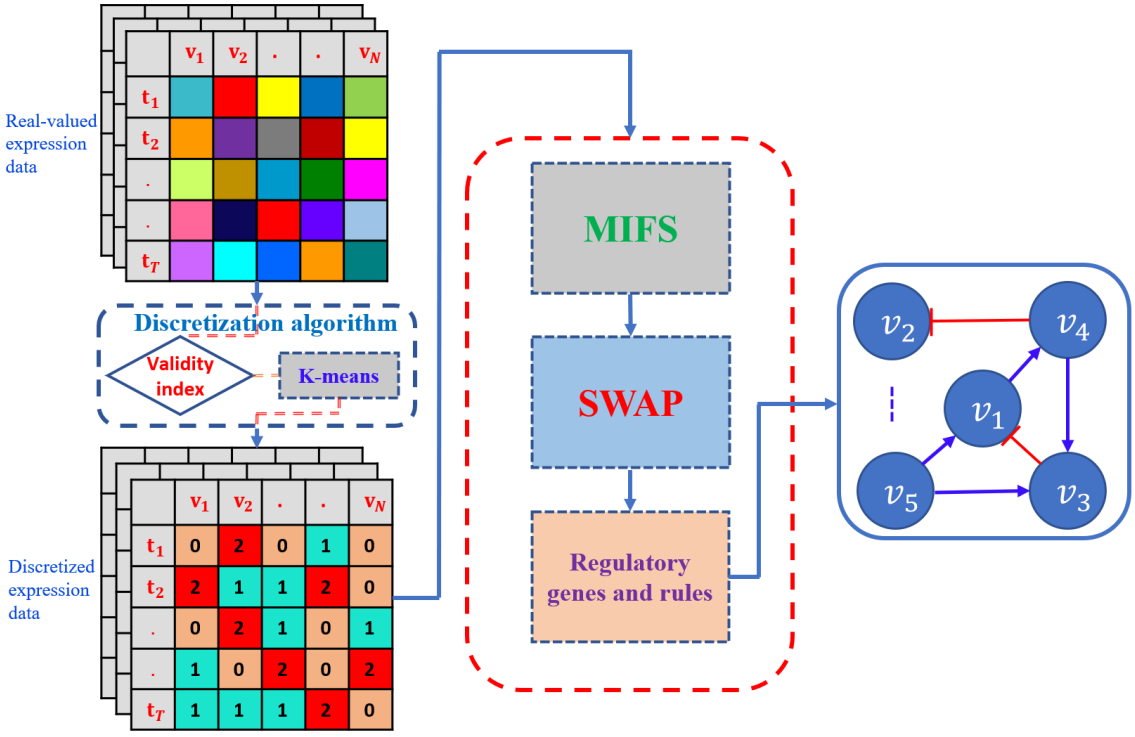


FIGURE 2.1: The overall framework of the MIDNI algorithm begins with a real-valued time-series gene expression dataset, which is transformed into a discretized expression dataset through the K-means discretization algorithm. The number of clusters K for each gene is adaptively determined using a validity index, ensuring that the discretization captures the inherent distribution of the gene's expression values. The discretized dataset is subsequently processed by the **MIFS** and **SWAP** subroutines, which identify the most informative regulatory genes for each target gene. The output of these subroutines is a collection of candidate regulatory sets for all genes in the network. Using these results, the gene regulatory network is reconstructed into a predictive network that can model and simulate the dynamic interactions among genes.

Discretization

Boolean discretization can significantly accelerate the network inference process, and it is well-suited for a subset of genes whose expression dynamics are sufficiently captured by two discrete states. Nevertheless, other genes exhibit more complex expression patterns that are better represented using a higher-level discretization, as illustrated in Figure 2.2. Indeed, previous studies have demonstrated the advantage of employing three-level discretization for gene expression data. For instance, Deep Neural Network approaches (Ahn et al., 2019) utilize the values 1, 0, and -1 to quantify individual gene contributions, reflecting upregulated, unchanged, and downregulated states. Similarly, several other methods (Ji and Kanehisa, 2004; Madeira and O., 2005; Gallo, J., and Ponzoni, 2013)

have adopted a ternary discretization scheme for genes, where 1, 0, and -1 correspond to "upregulated", "no-change", and "downregulated", providing a more nuanced representation of gene activity that captures intermediate regulatory effects.

Therefore, MIDNI adopts a hybrid discretization strategy, utilizing both binary (two-level) and ternary (three-level) representations. To select the optimal discretization level for each gene, the validity index is employed (Ray and Turi, 1999; Halkidi, Batistakis, and Vazirgiannis, 2001; Shen et al., 2005), which provides a quantitative measure of the clustering quality and is defined as follows:

$$\text{Validity} = \frac{\text{Intra}}{\text{Inter}}, \tag{2.1}$$

where

$$\text{Intra} = \frac{1}{P}\sum_{i=1}^{k}\sum_{x\in C_i}\|x - z_i\|^2, \tag{2.2}$$

and

$$\text{Inter} = \min_{i,j}\|z_i - z_j\|^2, \quad i \neq j \in \{1, 2, \ldots, k\}, \tag{2.3}$$

where P represents the total number of observed time points for the gene, k is the number of clusters, and z_i denotes the centroid of cluster C_i. The optimal number of clusters is then determined by selecting the value of k that minimizes the validity index, thereby ensuring an appropriate balance between cluster compactness and separation.

More specifically, each gene is discretized using K-means clustering with both two-class and three-class configurations. For each discretization scheme, the corresponding validity index is computed. The gene is then assigned to the discretization level (two or three classes) that yields the smaller validity index, indicating a better clustering structure. This procedure is applied independently to all genes in the network.

MIFS and SWAP subroutines

Given a discretized time-series gene expression dataset, *MIDNI* executes two subroutines, *MIFS* and *SWAP*, in sequence. We note that these subroutines were adapted from those of *MIBNI* (Barman and Y. K. Kwon, 2017), since the original versions were designed exclusively for binary-valued gene expression data.

For each target gene v in the network, the *MIFS* subroutine identifies an initial subset of regulatory candidates $S_v \subseteq V$ by evaluating an approximation of multivariate mutual information (see Algorithm 2.3). Afterward, the *SWAP* subroutine (see Algorithm 2.4) iteratively searches for an improved regulatory set beginning from S_v.

The function `search_update_rule` is responsible for identifying a set of update rules in the update table such that the regulatory set S_v can perfectly reproduce the discretized trajectory of the target gene v. An illustrative example of the `search_update_rule` function is provided in the Appendix.

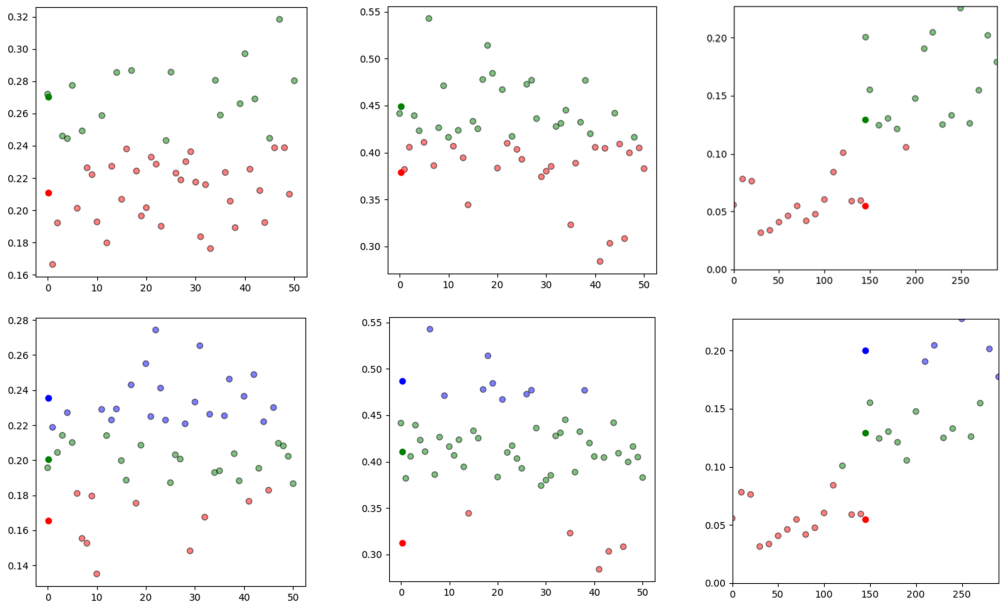


FIGURE 2.2: Illustrative examples of discretization for three different genes under two discretization scenarios. The upper row shows the discretized expression profiles obtained with $k = 2$, while the lower row presents the corresponding results with $k = 3$. For each gene, the x-axis represents the time points and the y-axis denotes the real-valued gene expression levels prior to discretization, highlighting how different discretization levels capture distinct expression patterns.

More specifically, the *SWAP* procedure repeatedly tests pairwise swaps between candidate regulatory genes in S_v (denoted as S in Algorithm 2.4) and genes in the complement set $V \setminus S_v$ (denoted as W). These swaps are tested exhaustively for all gene pairs across the two sets, following a decreasing order of mutual information with respect to the target gene.

As a final output, *MIDNI* produces the optimal regulatory gene set identified for each target gene in the network.

2.4 Experiments, Results, and Discussions

2.4.1 Experiments

The whole process can be divided into three main stages. In the first stage, the input gene expression data are discretized using the K-means clustering algorithm. The resulting discretized dataset is then processed by the *MIDNI* framework, whose output is a predicted regulatory structure of the underlying gene network. In the final stage, this inferred structure is evaluated by comparing it with the ground-truth network using tools that measure both structural and dynamical accuracies. All stages of the workflow are implemented using the Python programming language and its scientific libraries. To assess the performance and

robustness of *MIDNI*, we conducted experiments on two types of time-series gene expression datasets: an artificial discretized dataset and a real-valued dataset generated from Hill-function–based simulations.

2.4.2 Results

Case study 1: Artificial discretized dataset

We generated an artificial dataset consisting of 20 groups of discretized networks with sizes $|V| = 10, 20, \ldots, 190, 200$. Each group contains 20 independently generated random networks, resulting in a total of 400 networks used in this simulation study. For each target gene, the regulatory structure was created by randomly selecting k regulators from all genes except the target gene itself. The value of k was also sampled uniformly at random from the interval [1, maximum number of regulators allowed] for each target gene (see Table 2.1).

TABLE 2.1: The settings used for generating the datasets.

Parameters	Discretized dataset	Real-valued dataset
Size of networks (N)	10, 20, ..., 190, 200	50, 100, 200, 300
Noise rate	None	0, 5, 10
Time lag	1	1
Number of observed time points	30	30
Number of regulators in the network	$0.4 \times N$	$0.4 \times N$
Maximum regulators for each gene	4–8	4–8
Max expression ($\alpha_{\max}$)	NA	0.1–1
Basal expression (β_0)	NA	0.1–1
Apparent dissociation constant (K_d)	NA	1–5
Hill coefficient (n)	NA	1–5

In this simulation setting, the initial state of each gene was randomly assigned one of the three discrete values 0, 1, or 2. The state of every node was then updated across 29 consecutive time-steps using a discrete update function, which was chosen uniformly at random from the full set of possible functions. For a target gene regulated by k genes, the number of possible update functions is $3^{(3^k)}$. Based on this procedure, we generated the artificial discretized time-series dataset without requiring any additional discretization step (the detailed parameter configuration is summarized in Table 2.1).

The inference results obtained by MIDNI are shown in Figure 2.5. As illustrated, the dynamics accuracy was close to 1.0 across all network sizes. This near-perfect performance is likely related to the constraint that each gene has at most eight regulatory inputs. Regarding structural metrics, precision and recall gradually decreased as the network size increased, whereas the structural accuracy increased. The latter trend is expected because larger networks contain more candidate non-regulatory pairs, leading to an increase in true negative cases.

To further assess the robustness of MIDNI, we evaluated how the number of incoming regulatory links (D) affects inference quality (Figure 2.6). In the gold-standard networks, D represents the true number of regulators for a given target gene and reflects the intrinsic difficulty of the inference task. The generated

networks exhibited $D \in [1, 8]$. When $D = 1$ or $D = 2$, all performance indicators—both dynamics and structure-based—were nearly perfect. For $D > 2$, the structural metrics progressively declined, although the dynamics accuracy remained close to 1.0 even at $D = 8$.

These observations highlight the multimodal nature of the GRN inference problem: many different network structures can produce nearly identical dynamical behaviors. Consequently, MIDNI consistently identifies networks that replicate the temporal dynamics of the gold-standard networks, whereas perfect structural recovery becomes inherently challenging due to this multimodality.

Case study 2: Artificial real-valued dataset

MIDNI was further evaluated using a real-valued time-series gene expression dataset generated based on the Hill function (Santillan, 2008). This dataset contains four groups of networks with sizes $|V| = 50, 100, 200$, and 300. Each group includes 20 independently generated random networks, resulting in a total of 80 networks for this experiment. The procedure for constructing the network topology followed the same approach used for the artificial discretized dataset. The Hill function employed to simulate the continuous gene expression dynamics is given below, and its corresponding parameter settings are summarized in Table 2.1.

$$[\text{Protein}] = \begin{cases} \alpha_{\max}\left(\beta_0 + (1-\beta_0)\dfrac{[\text{TF}]^n}{K_d + [\text{TF}]^n}\right), & \text{for activation,} \\ \alpha_{\max}\left(\beta_0 + (1-\beta_0)\dfrac{K_d}{K_d + [\text{TF}]^n}\right), & \text{for inhibition,} \end{cases}$$

where $\alpha_{\max}$, β_0, TF, K_d, and n represent the maximum transcriptional output, the basal expression level, the transcription factors regulating the gene, the apparent dissociation constant, and the Hill coefficient, respectively. As described in Section 3.2.1, the expression profile of each gene was discretized using K-means clustering, where the number of clusters k was chosen to be either two or three depending on the underlying distribution of the gene's expression values.

We first assessed the proportion of genes assigned to the three-level discretization category, as illustrated in Figure 2.7. Except for the networks of size 10, the proportions of genes discretized into two levels and those discretized into three levels were approximately balanced. This observation indicates that ternary discretization occurs as frequently as binary discretization within our framework.

To further evaluate performance, we compared MIDNI against several representative inference algorithms—DBN, MICRAT, MIBNI, and GENIE3—using artificial time-series gene expression datasets. DBN (Zou and Conzen, 2005) identifies potential regulators by analyzing the earliest time points at which gene expression exhibits up- or down-regulation. MICRAT (Yang et al., 2018) constructs an undirected association graph based on the maximal information coefficient, and subsequently assigns edge directions using average entropy. GENIE3 (Vân Anh Huynh-Thu et al., 2010), a tree-based inference approach, was the top-performing method in the DREAM4 In Silico Multifactorial Challenge. MIBNI represents our earlier Boolean-based method for network reconstruction.

Structural accuracy analysis

We grouped all target genes into eight categories based on the number of incoming regulatory links (D), which varies from 1 to 8, as illustrated in Figure 2.8. As shown in the figure, the structural accuracy of all inference methods consistently declined as D increased. This trend occurs because a larger number of incoming links makes the inference task inherently more challenging. When $D = 1$, nearly all methods successfully identified the optimal solution. However, as D grew, their performance decreased in an approximately linear manner.

We emphasize that our proposed method achieved the highest structural accuracy in every setting, except for the cases $D = 1$ and $D = 2$ in the smallest network ($|V| = 50$). The performance advantage became more pronounced when both the network size and the value of D were large. These findings demonstrate the effectiveness of our approach in large-scale and complex inference scenarios. Furthermore, the integration of multi-level discretization provided additional improvement compared with our previous method, MIBNI.

Dynamics accuracy analysis

We recognize that distinct network structures can sometimes lead to the same underlying dynamics. Therefore, it is also important to evaluate the inference performance in terms of dynamics accuracy. We assessed the dynamics accuracies of the inferred networks generated by MIDNI, DBN, MICRAT, MIBNI, and GENIE3. Similar to the structural accuracy analysis, we examined how the number of regulatory genes for each target gene (D) influences the dynamics accuracy of each method. The corresponding results are presented in Figure 2.9.

As shown in the figure, most methods achieved dynamics accuracy close to 1.0 when $D = 1$. As D increases, the inference task becomes more challenging, resulting in lower dynamics accuracies. However, MIDNI exhibited the best performance in all cases except for $D = 3$ in the network group with size $|V| = 300$. In particular, for $D = 8$ in the network groups of sizes 200 and 300, MIDNI achieved a dynamics accuracy of approximately 0.6, while the other methods remained around 0.5. These results highlight the strong inference capability of our method for more complex and larger-scale networks.

Running time

To assess the computational efficiency of MIDNI in comparison with other inference methods, we evaluated the average running time over the 80 random networks described in Case Study 2. All experiments were performed on a PC equipped with an AMD Ryzen 5 3400G 3.70 GHz CPU and 16 GB RAM. The results are illustrated in Figure 2.10, where the y-axis indicates the average running time in milliseconds for the four different network sizes.

Although MIDNI was noticeably slower than DBN and MICRAT, its running time was comparable to that of GENIE3 and MIBNI, and it was slightly faster than MIBNI. This improvement is due to several optimizations applied to the MIFS and SWAP subroutines in MIDNI compared to the original implementation. For instance, we employed built-in libraries such as `sklearn` and `scipy` to compute mutual information and entropy metrics, which are more efficient than custom-coded versions. We also utilized vectorized computations, which execute significantly faster than traditional for-loop based calculations.

Considering that MIDNI achieved superior structural and dynamics inference accuracy compared with the other methods, it is a desirable choice when high accuracy is required, even at the expense of additional computational time.

2.4.3 Discussions

In this study, we proposed a new network inference method for time-series gene expression data, called MIDNI. The method first converts real-valued expression profiles into multi-level discretized values according to the underlying distribution of the expression data. Unlike most previous discretization-based inference approaches that employ Boolean network models, which may limit inference accuracy due to the oversimplified representation, MIDNI achieved superior performance in terms of both structural and dynamical accuracy, particularly in larger and more densely connected networks.

Despite these promising results, MIDNI has several limitations. First, the precision and recall values were not sufficiently high. We attribute this to the reliance on mutual information for selecting candidate regulatory genes that directly interact with a target gene. Some regulatory effects may be mediated through intermediate genes, which can lead to incorrect candidate selection. This suggests that integrating prior biological knowledge from existing databases may lead to more accurate predictions. Second, MIDNI uses the correlation coefficient to determine the interaction type and direction, which may also contribute to reduced accuracy. Finally, the MIFS and SWAP subroutines employed in MIDNI are greedy algorithms, and the inference performance could potentially be improved by incorporating more advanced or global search strategies.

2.5 Conclusion & Future Studies

2.5.1 Conclusion

This study addresses one of the fundamental and challenging problems in computational biology, namely the inference of regulatory gene networks. The primary distinction of the proposed method compared with existing approaches lies in its use of a multi-level data representation scheme rather than relying on raw real-valued expression data or Boolean discretization. This choice not only prevents information loss during representation but also accelerates the inference process. Consequently, the method achieves superior structural and dynamical accuracy, particularly for large-scale networks.

Several key issues in network inference have been effectively addressed by the proposed approach:

- **Adaptive discretization.** MIDNI determines the optimal number of clusters (i.e., discretization levels) for each gene based on its expression value distribution.

- **Multivariate candidate evaluation.** The algorithm identifies plausible regulatory genes using not only pairwise mutual information but also the approximate dependency of a candidate on the set of previously selected regulators. This multivariate assessment distinguishes MIDNI from most existing methods that consider only pairwise interactions.
- **Enhanced structural and dynamical accuracy.** MIDNI improves not only the structural agreement between inferred and gold-standard networks but also the dynamical consistency of gene expression patterns within the network.

2.5.2 Future Studies

Although the proposed method outperformed other compared approaches, it still has several limitations, as discussed in subsection 2.4.2, which should be addressed in the next stage of this research. In the future, I plan to extend this study in several directions:

- Develop a new subroutine capable of handling multimodal inference problems, in which many optimal solutions may exist.
- Apply more sophisticated metrics to determine the direction of gene interactions during inference, since the correlation coefficient is relatively simple and may not sufficiently improve accuracy.
- Accelerate the inference process by integrating optimized search algorithms or parallel computation techniques.
- Conduct experiments on data representation using alternative algorithms to compare with the current K-means-based discretization.
- Attempt to infer networks using other types of data, such as steady-state expression profiles (since this study used only time-series data), and incorporate prior biological knowledge to further enhance inference accuracy.

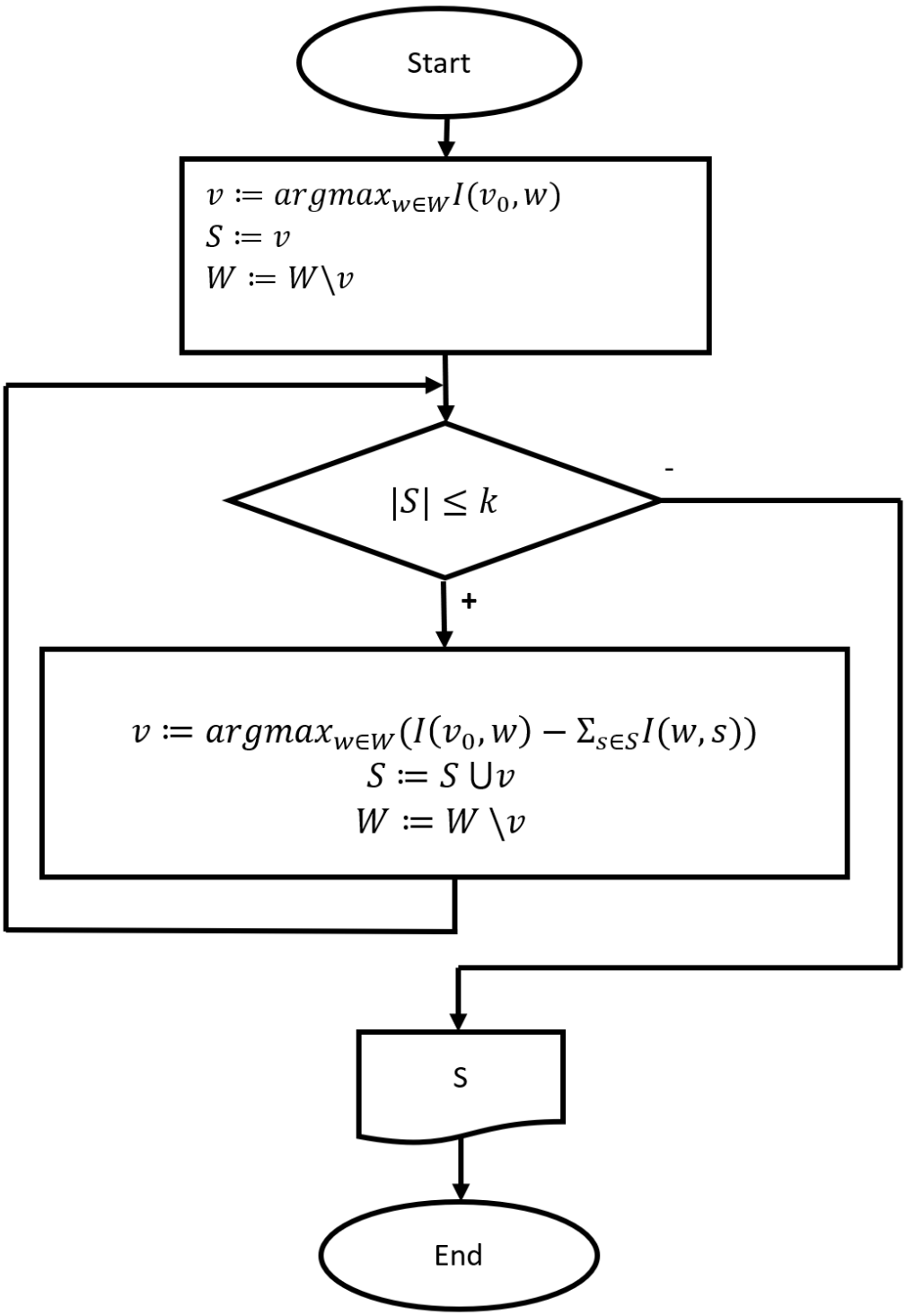


FIGURE 2.3: The subroutine $MIFS(v_0, W, k)$, where v_0 is the target variable, $W = \{w_1, w_2, \ldots, w_M\}$ is the set of variables, and k is the desired number of input variables.

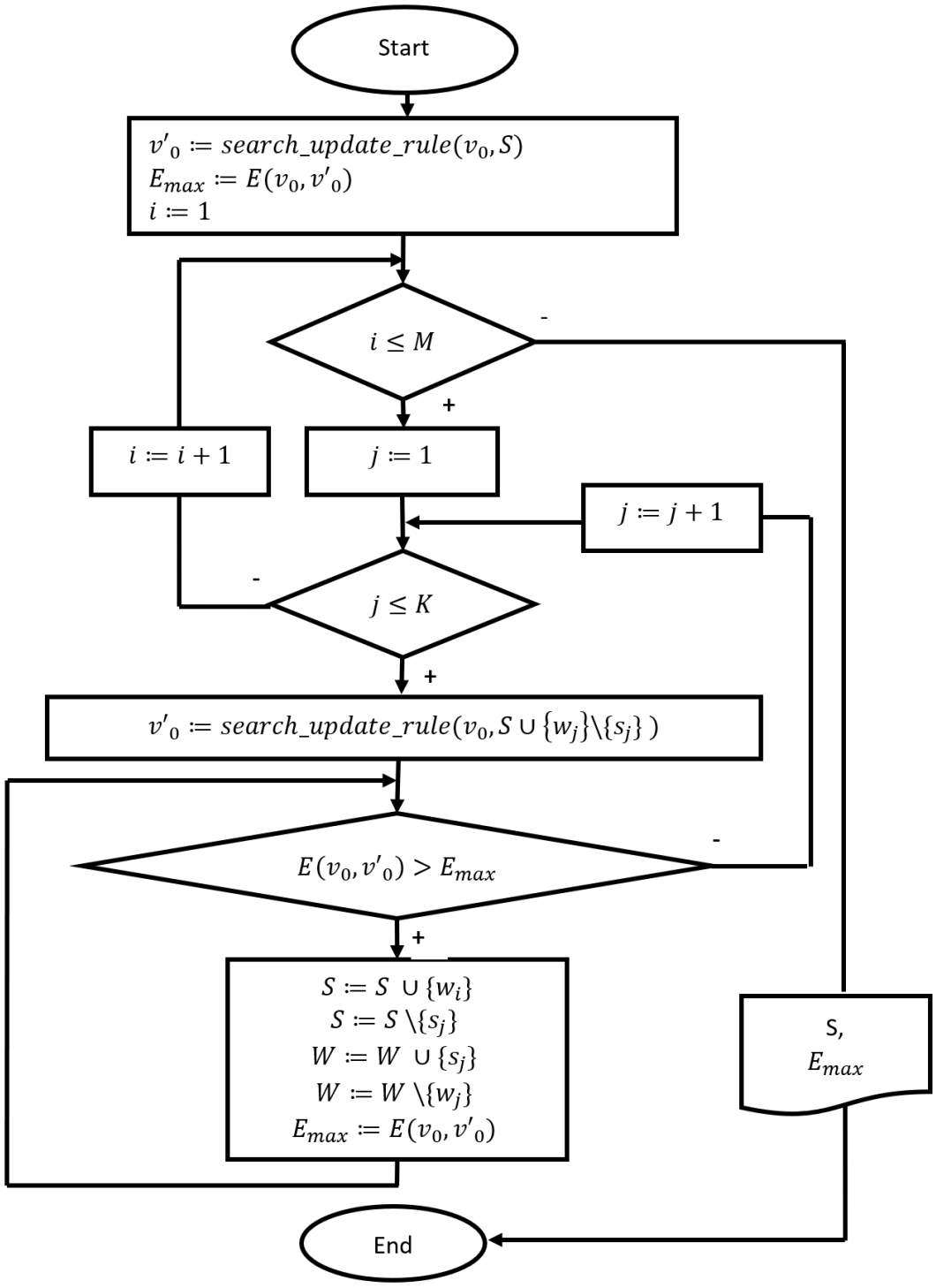


FIGURE 2.4: The subroutine $SWAP(v_0, S, W)$, where v_0 is the target variable; $S = \{s_1, s_2, \ldots, s_k\}$ is the set of selected variables such that $I(v_0, s_i) \geq I(v_0, s_j)$ if $i < j$ for all $s_i, s_j \in S$; and $W = \{w_1, w_2, \ldots, w_M\}$ is the set of unselected variables such that $I(v_0, w_i) \geq I(v_0, w_j)$ if $i < j$ for all $w_i, w_j \in W$.

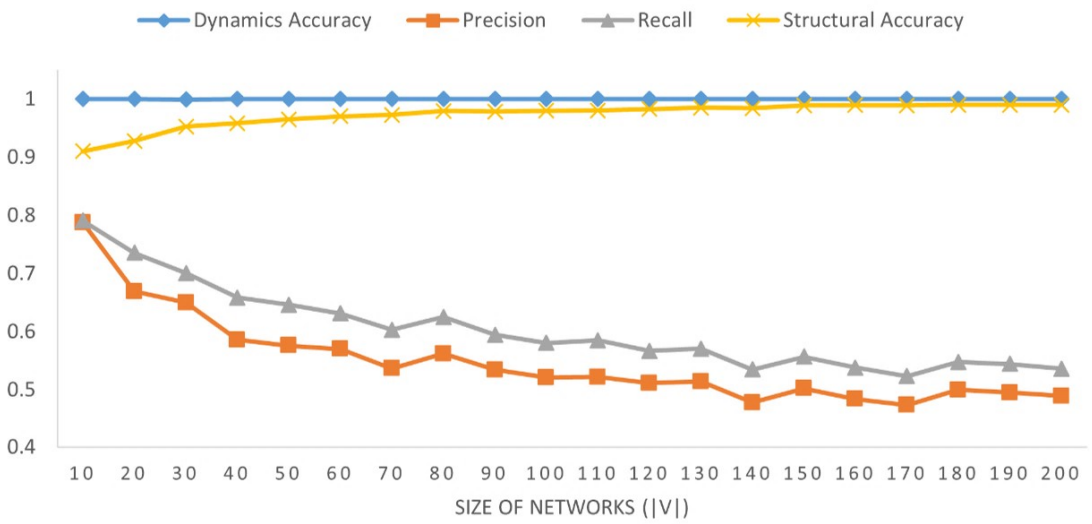


FIGURE 2.5: The performance of MIDNI on the artificial discretized dataset. The x-axis means the network size denoted by the number of genes in the network. For each size, the y-axis shows the average value of each performance metric over 20 random networks.

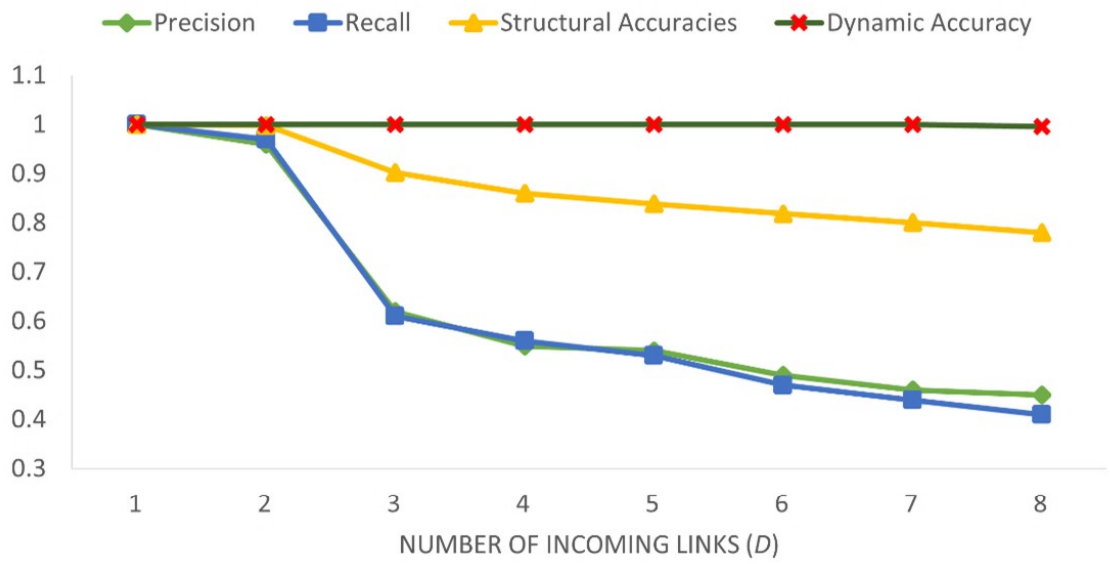


FIGURE 2.6: Inference performance according to the number of incoming links (D) in the gold-standard network.

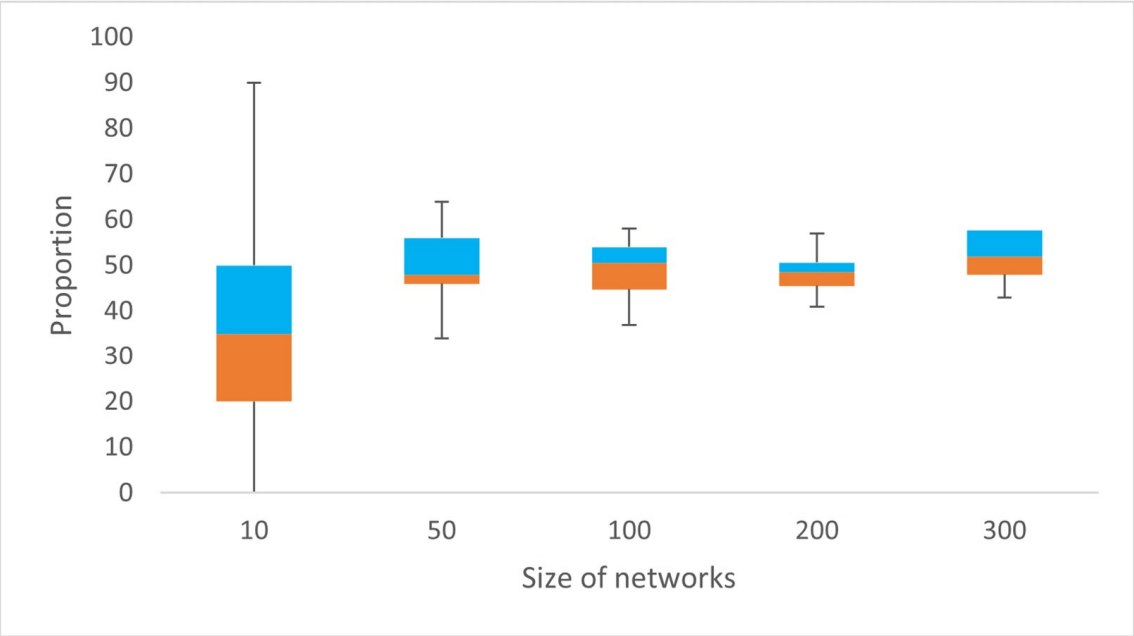


FIGURE 2.7: The proportion of three-level discretized genes in the networks.

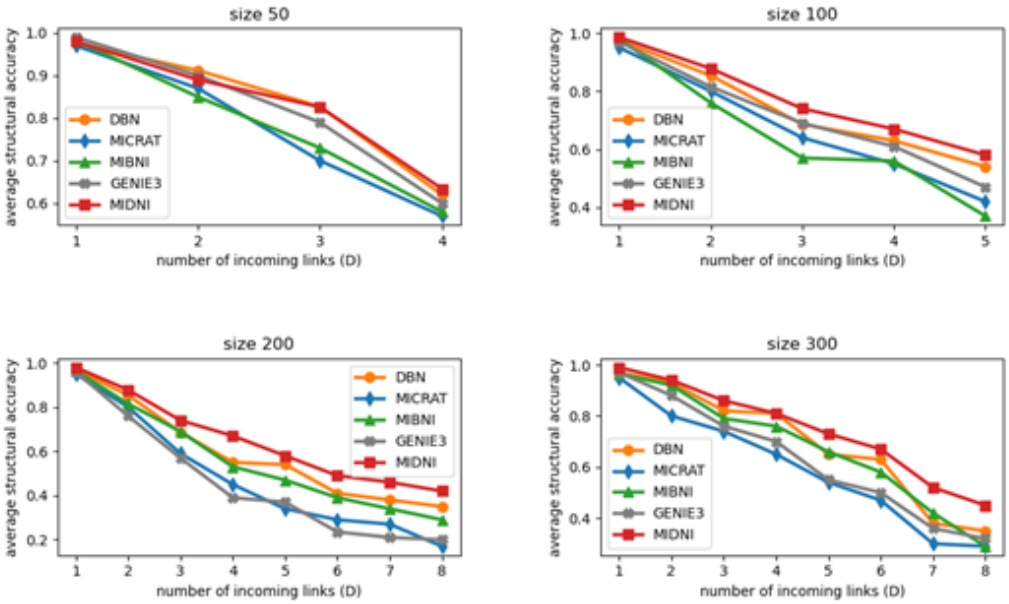


FIGURE 2.8: Structural performance on random real-valued networks. Four network size groups ($|V| = 50, 100, 200, 300$) were examined, with 20 randomly generated networks for each size. Target genes were categorized based on the number of incoming regulatory links (D) in the gold-standard network. The x-axis represents the values of D, while the y-axis indicates the average structural accuracy across networks.

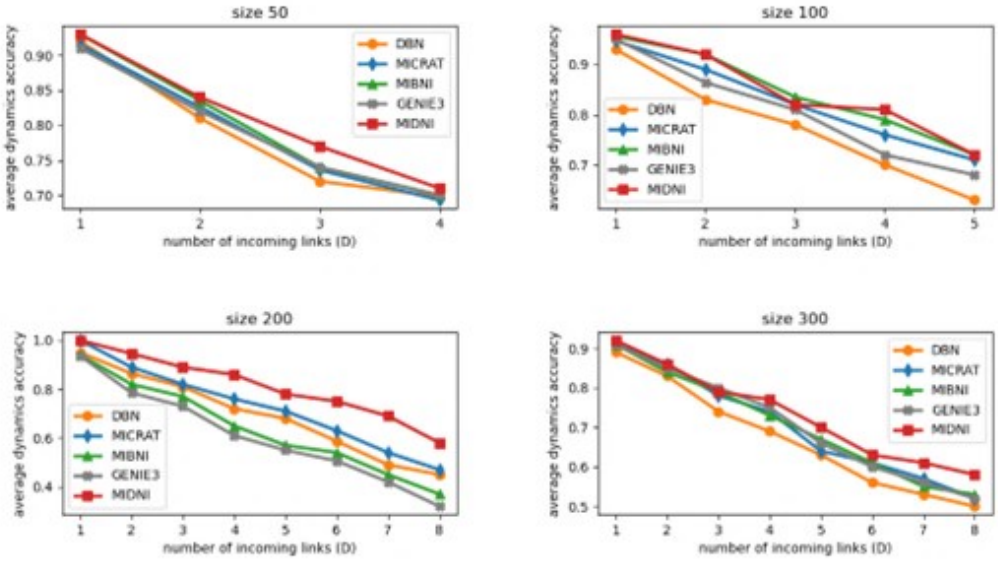


FIGURE 2.9: Dynamical performance on the random real-valued networks. Four size groups of networks ($|V| = 50, 100, 200$, and 300) were considered, and 20 random networks were generated for each group. All target genes are grouped according to the number of incoming links (D) in the gold-standard network, and the x-axis represents the D values. The y-axis shows the average dynamics accuracy.

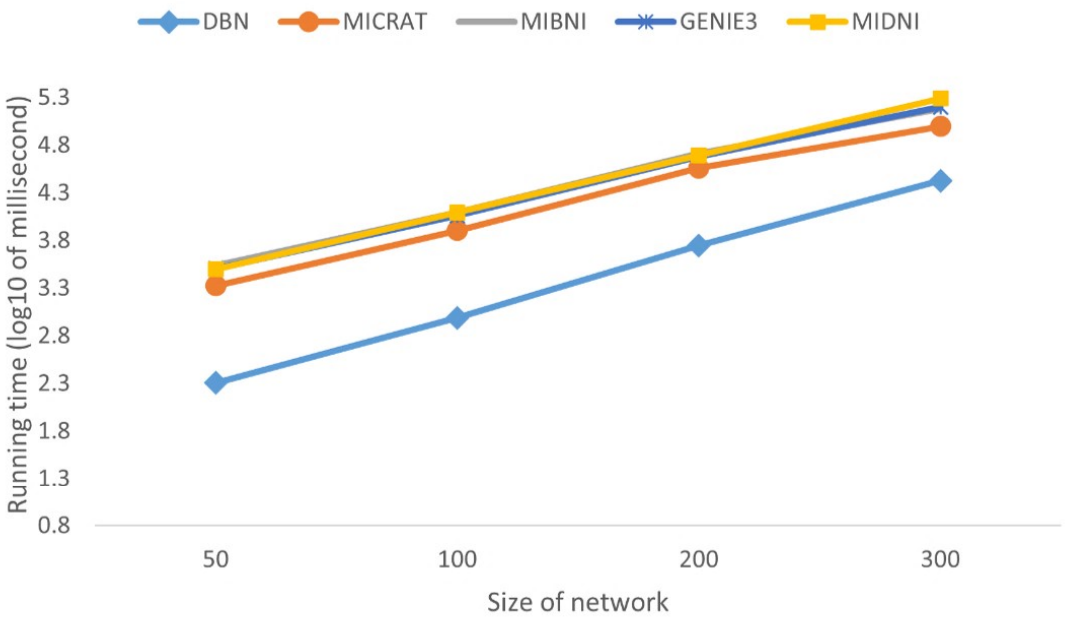


FIGURE 2.10: Comparison of running time across different network sizes. The x-axis represents the network size, while the y-axis shows the running time in milliseconds on a $\log_{10}$ scale.

Chapter 3

A Representation Learning Framework for Gene Regulatory Network Inference from Time-series Expression Data

3.1 Abstract

GRNs describe the complex regulatory relationships that govern gene expression and play a central role in fundamental biological processes such as development, cellular differentiation, and responses to environmental perturbations. Inferring GRNs from time-series gene expression data is therefore a key problem in computational biology. However, this task remains challenging due to the high dimensionality of expression profiles, limited sample sizes, intrinsic noise, and the dynamic and nonlinear nature of gene regulation. Classical GRN inference methods typically rely on explicit statistical or mechanistic models, such as correlation analysis, regression frameworks, mutual information estimation, or Bayesian formulations. While these approaches have provided important insights, they often depend on strong assumptions regarding linearity, time delays, or the number of candidate regulators, which can limit robustness and generalizability across datasets.

In this thesis, we propose a novel data representation framework that reformulates GRN inference as a binary classification problem over gene pairs. Instead of explicitly modeling regulatory dynamics through predefined equations or probabilistic structures, the proposed framework encodes each potential regulatory interaction into a fixed-dimensional feature vector that jointly captures temporal expression dynamics and local structural context. This representation enables the direct application of a wide range of supervised machine learning models, allowing regulatory relationships to be inferred in a data-driven and assumption-free manner. By avoiding heuristic parameters such as fixed time lags or predefined regulator sets, the framework improves flexibility and reduces sensitivity to dataset-specific tuning.

A key property of the proposed representation is its independence from the global size of the underlying network. Because regulatory interactions are encoded locally, interactions drawn from networks of different sizes share a consistent representation. This scale invariance facilitates cross-network generalization

and enables training data to be constructed from mixtures of networks with varying sizes, which is particularly beneficial for mitigating class imbalance between interacting and non-interacting gene pairs. Moreover, once trained, the resulting classifiers support efficient inference, as models can be trained once and reused to predict regulatory interactions across large collections of networks.

We evaluate the proposed approach on benchmark synthetic datasets generated from the *Escherichia coli (E.coli)* regulatory network topology, considering both discrete-valued toy datasets and continuous-valued time series data, and across multiple network sizes. Experimental results demonstrate that the proposed framework achieves consistently strong performance across different data generation regimes and network scales, outperforming traditional GRN inference approaches that rely on explicit statistical or model-based assumptions. These findings highlight the robustness, scalability, and practical applicability of the proposed representation learning framework.

Overall, this work demonstrates that reframing GRN inference through a principled data representation and supervised learning paradigm provides a flexible and scalable alternative to traditional modeling-driven approaches. The proposed framework offers a general foundation for integrating modern machine learning techniques into GRN reconstruction and holds promise for future applications in systems biology and large scale genomic data analysis.

3.2 Introduction

GRNs describe the complex and coordinated interactions among genes and their regulatory elements, which collectively govern essential biological processes such as cell differentiation, development, and cellular responses to environmental stimuli. Accurate inference of GRNs from time-series gene expression data is therefore a fundamental problem in computational biology, as it provides mechanistic insights into gene regulation and supports downstream applications in disease modeling, precision medicine, synthetic biology, and drug discovery. However, reconstructing GRNs from expression data remains challenging due to the high dimensionality of genomic measurements, the presence of stochastic noise, and the limited number of temporal samples typically available in experimental settings.

A wide range of computational approaches has been proposed for GRN inference, spanning from classical statistical techniques to more advanced machine learning models. Early correlation based methods, such as Pearson and Spearman correlation, infer regulatory relationships by measuring pairwise linear or monotonic associations between gene expression profiles (Langfelder and Steve Horvath, 2008; Guo et al., 2014). Although computationally efficient, these approaches fail to capture complex nonlinear dependencies and are highly sensitive to noise, which often results in inflated false positive rates. Regression based methods, including GENIE3 (Vân Anh Huynh-Thu et al., 2010), fused LASSO models (Omranian et al., 2016), and KFLR (Pirgazi and Khanteymoori, 2018), extend this paradigm by modeling gene expression as a function of potential regulators while imposing sparsity constraints to reduce spurious interactions. Despite improved specificity, these methods rely on explicit functional assumptions, such

as linearity or tree-based relationships, and typically require large sample sizes to achieve stable and reliable inference.

Mutual information based approaches, such as ARACNE (Margolin et al., 2006), CLR (Faith et al., 2007), and MIDNI (Anh and Y.-K. Kwon, 2023), relax linearity assumptions by quantifying general statistical dependencies between gene pairs. While these methods are more effective at detecting nonlinear interactions and exhibit improved robustness to noise, they often depend on discretization of continuous expression values, which can lead to information loss, and they face difficulties in distinguishing direct regulatory effects from indirect associations. Probabilistic frameworks, including Bayesian networks (Wilczyński and Dojer, 2009) and dynamic Bayesian networks (Ajmal and Madden, 2021; Suter, Kuipers, and Beerenwinkel, 2022), further incorporate temporal dependencies and uncertainty modeling. Although theoretically well grounded, these approaches are computationally demanding and sensitive to prior assumptions, which limits their scalability to large networks and long time-series datasets.

More recently, task transformation or problem reformulation, has emerged as a promising strategy for addressing the complexity of GRN inference. Instead of explicitly modeling regulatory functions or dependency structures, several methods recast GRN reconstruction into supervised learning tasks that are more amenable to modern machine learning algorithms. For example, CNNGRN (Gao et al., 2023) reformulates GRN inference as a binary classification problem by constructing joint representations of gene expression profiles and network features, enabling convolutional neural networks to capture spatial and temporal patterns. Similarly, MKL-GRNI (Wani and Raza, 2021) adopts a supervised classification framework at the transcription factor level and integrates multi-omics data through multiple kernel learning to improve robustness against noise and high dimensionality. GRNBoost2 (Moerman et al., 2019) reframes GRN inference as a regression based ranking task, which is subsequently converted into a binary classification problem to identify high confidence regulatory interactions, leveraging gradient boosting to scale to large single cell datasets.

Despite their demonstrated effectiveness, these task transformation based approaches share several common limitations that restrict their general applicability. In particular, they often rely on complex or method specific data representations that are difficult to interpret biologically and may not generalize well across different network sizes or experimental conditions. Moreover, temporal expression dynamics and regulatory structural context are typically encoded separately or indirectly, which limits the ability of the models to jointly capture complementary aspects of gene regulation. In addition, many existing formulations depend on predefined assumptions, such as fixed time delays, selected candidate regulators, or kernel configurations, which can constrain flexibility and hinder robustness when applied to heterogeneous datasets.

Motivated by these shared limitations, we propose a novel data representation framework that reformulates GRN inference as a binary classification problem through a unified and compact encoding of gene pair relationships. By jointly representing time-series expression dynamics and regulatory structural information within a standardized vector space, the proposed framework eliminates the need for explicit assumptions regarding time lags or regulator selection, thereby

enabling assumption-free inference. This representation facilitates the direct application of a broad class of supervised learning models that have been extensively validated in other high-dimensional domains, allowing the proposed approach to leverage their robustness, scalability, and predictive power. Furthermore, the standardized nature of the representation enables seamless integration of data across networks of varying sizes, which supports cross-network learning and alleviates class imbalance, a persistent challenge in supervised GRN inference.

Empirical evaluations on benchmark GRN datasets demonstrate that the proposed method consistently outperforms existing approaches in terms of predictive accuracy and computational efficiency. These results highlight the potential of the proposed task transformation framework to overcome key limitations of both traditional and recent machine learning based GRN inference methods, providing a scalable, flexible, and biologically meaningful solution for reconstructing gene regulatory networks from time-series expression data.

3.3 Materials and methods

3.3.1 Related works

To evaluate the proposed method, which reformulates GRN inference as a binary classification task, we compare it with representative approaches spanning major paradigms for time-series GRN inference, including tree based, regression based, mutual information based, and Bayesian model based methods. Tree based methods are represented by JUMP3 (V. A. Huynh-Thu and Sanguinetti, 2015), which combines decision trees with dynamic programming to infer regulatory interactions from time-series data. Regression based approaches are represented by Inferelator (Greenfield et al., 2010) and TSNI (Bansal, Gatta, and Di Bernardo, 2006), which model gene expression dynamics using linear regression or linear differential equation frameworks with time-lagged predictors. Mutual information based methods are represented by CLR-lag (Faith et al., 2007), which extends the CLR framework by incorporating temporal delays to capture dynamic regulatory relationships. Bayesian model based approaches are represented by G1DBN (Lèbre, 2007), a Dynamic Bayesian Network method that explicitly models temporal conditional dependencies in gene regulation from time-series data.

3.3.2 Performance Metrics

In this study, we employ the Precision-Recall (PR) curve and its corresponding Area Under the Precision-Recall Curve (AUPR) as the primary evaluation metrics to assess the performance of gene network inference. The PR curve plots the trade-off between precision (the proportion of true positive predictions among all positive predictions) and recall (the proportion of true positives detected among all actual positives) across various classification thresholds. Unlike the ROC curve, which may present an overly optimistic view in imbalanced datasets, the PR curve is more informative in scenarios where the positive class (true gene-gene interactions) is significantly outnumbered by the negative class (non-interacting

pairs), as commonly seen in biological networks. The AUPR summarizes the PR curve into a single scalar value between 0 and 1, where a value of 1 indicates perfect classification and a value close to 0 suggests random or poor predictive performance. Therefore, AUPR serves as a robust indicator of model effectiveness in the context of sparse and noisy biological data.

3.4 Our proposed model

3.4.1 Data representation

Inferring gene regulatory networks from time series expression data remains challenging due to the complex interplay between temporal dynamics and network structure, as well as the severe class imbalance between interacting and non-interacting gene pairs. Traditional approaches often rely on handcrafted assumptions, such as fixed time delays or predefined regulator sets, which can limit their generalizability across datasets and network scales. To address these limitations, we seek a representation that directly encodes dynamic expression information at the level of individual gene pairs, while remaining agnostic to network size and structural priors. Such a representation should enable the application of supervised learning techniques by transforming GRN inference into a standard classification problem, thereby leveraging advances in machine learning for scalable and robust network reconstruction. Motivated by these considerations, we introduce a novel representation of time series gene expression data that supports supervised learning–based GRN inference. Instead of representing gene interactions using gene identifiers, we encode each pairwise interaction by concatenating their corresponding gene expression vectors over time.

Given a time series data set with T observed time points, the expression profile of a gene G_i is denoted by a vector:

$$\mathbf{V}_i = \{v_i^{(1)}, v_i^{(2)}, \ldots, v_i^{(T)}\}$$

where $v_i^{(t)}$ represents the expression value of gene G_i at time point t.

Each regulatory interaction in the gold standard file is given in the form of a directed edge $G_j \to G_i$, labeled 1 if an interaction exists and 0 otherwise. For each such gene pair, we construct a feature vector by concatenating the expression vectors of the regulator and the target:

$$\mathbf{x}_{(j,i)} = \mathbf{V}_j \oplus \mathbf{V}_i \in \mathbb{R}^{2T}$$

with the corresponding binary label:

$$y_{(j,i)} = \begin{cases} 1, & \text{if } G_j \to G_i \text{ exists} \\ 0, & \text{otherwise} \end{cases}$$

By applying this transformation across all gene pairs in the gold standard, we obtain a labeled dataset $\mathcal{D} = \{(\mathbf{x}_{(j,i)}, y_{(j,i)})\}$, where each sample represents a regulatory interaction.

This representation framework confers several notable advantages for GRN inference. First, by integrating both the structural context of regulatory interactions and the temporal dynamics of gene expression, it yields a more informative feature space that captures complementary aspects of regulatory behavior. This joint encoding enhances the model's capacity to detect true interactions, outperforming approaches that rely solely on temporal expression profiles. Second, the method operates in an assumption-free manner, obviating the need for predefined parameters, such as time delays or the number of candidate regulators, requirements that often constrain traditional inference techniques. Specifically, temporal dependencies are implicitly captured through the encoded expression dynamics rather than being imposed via fixed time-lag assumptions, while regulatory candidates are learned directly from data without restricting the search space to a predefined subset. Third, the reformulation of the GRN reconstruction problem as a binary classification task enables the direct application of a broad class of supervised learning models, facilitating robust and scalable prediction. Finally, the standardized vector representation allows for seamless integration of data across networks of varying sizes, thereby supporting cross network generalization and mitigating class imbalance which is an inherent difficulty in supervised GRN inference tasks. In large networks (e.g., size 100), true regulatory interactions typically constitute only a small fraction of all possible gene pairs, leading to severe imbalance between interaction and non-interaction classes. Under this unified representation, interaction samples derived from networks of smaller sizes (e.g., size 10 or size 50) can be naturally incorporated into the training set to enrich positive examples. This strategy is feasible because all gene pairs are encoded into fixed-length feature vectors with identical dimensionality, regardless of network size, enabling effective knowledge transfer across datasets and improving model robustness in large scale settings.

3.4.2 The model

The Figure 3.1 presents the entire framework of our proposed approach. It begins with a collection of gold standard gene regulatory network structures that specify the true regulatory interactions between gene pairs. These networks provide the structural foundation from which simulated expression dynamics are produced. Time-series expression profiles for every network structure are generated using the GeneNetWeaver (GNW) simulator. Each profile contains the expression trajectories of all genes across T time steps. These expression profile is then converted into a supervised learning dataset using the representation procedure described in 3.4.1. This procedure transforms raw temporal expression values into feature and label pairs that encode the local regulatory context of each candidate interaction. The resulting represented dataset forms a unified feature space that is compatible with a wide variety of machine learning algorithms that include linear methods, tree based approaches, ensemble techniques, probabilistic models, non parametric classifiers, and neural architectures. In particular, we employ Linear Regression (LR), Decision Tree (DT), Random Forest (RF), Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), Gaussian Naive Bayes (GNB), K Nearest Neighbors (KNN), and a Multi Layer Perceptron

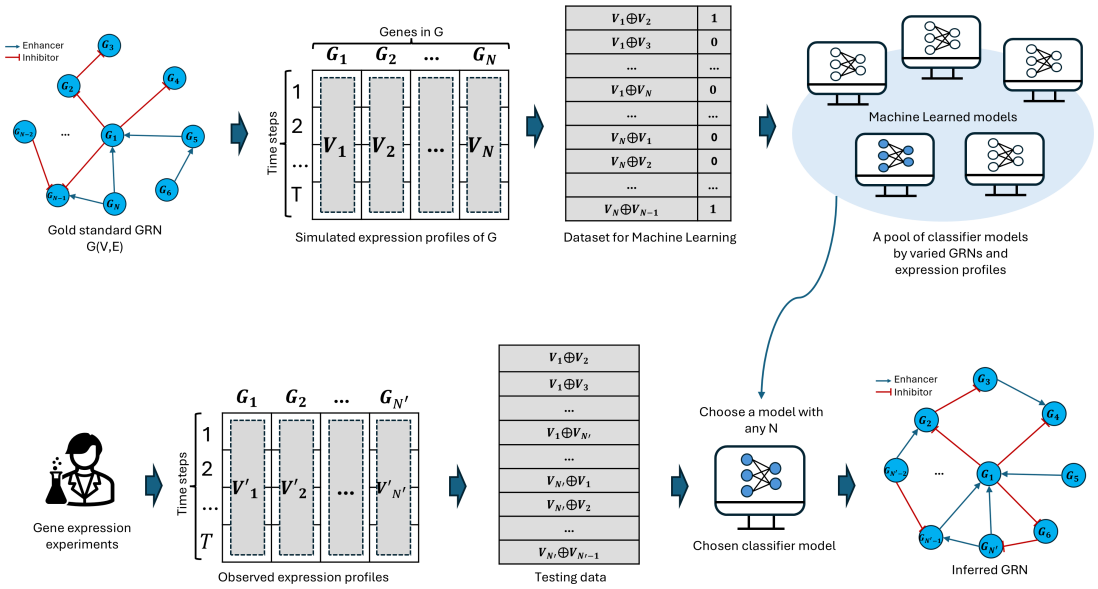


FIGURE 3.1: Overview of the proposed GRN inference framework. The key required input is the observed number of timesteps, which determines how simulated trajectories are aligned with the test instances. In contrast, network size is not a constraining factor, since the proposed representation enables models trained on networks of one size to generalize reliably to networks of different sizes. The framework produces pairwise interaction predictions for all gene pairs, which are then aggregated to reconstruct the inferred regulatory network.

(MLP). (MLP).This comprehensive model set ensures that the evaluation reflects not only the effectiveness of the proposed data representation but also its general applicability across fundamentally different learning mechanisms. The parameter configurations applied to these models during the training process are detailed in Table 3.1. After training, all models are stored in a pool of classifier models, which allows users to select a preferred model when performing inference on new data. When presented with an unseen expression dataset, the inference pipeline assumes that the data contain exactly T observed time steps, matching the temporal length used during model training. Under this setting, the unseen expression profiles are directly transformed using the same representation strategy without requiring any additional preprocessing or temporal alignment. The number of observed time steps therefore constitutes the primary prerequisite for inference, and this parameter is typically well defined for most experimental time series datasets. Once the expression data are transformed into the represented feature space, a selected model from the classifier pool is applied to predict interaction labels for all candidate gene pairs. The final output of the inference pipeline is a binary label for each gene pair indicating interaction or non interaction. These pairwise labels are then assembled to reconstruct the inferred gene regulatory network.

TABLE 3.1: The hyper parameters of classifiers

Model name	Parameter Name	Value
Linear Regression Gaussian Naive Bayes Decision Tree	Default setting	
Random Forest	Number of estimators	20
KNN	Number of neighbors	10
XGBoost	Max-depth	3
	Learning rate (eta)	0.1
	Number of boosting rounds	100
	Early stopping rounds	10
LightGBM	Boosting type	gbdt
	Leaves number	32
	Learning rate	0.05
	Feature fraction	0.9
	Number of boosting rounds	100
MLP	Number of hidden layers	5
	Activation functions in hidden layers	relu
	Activation function in the output layer	sigmoid
	Optimizer	Adam
	Learning rate	0.01
	Loss function	mse
	Early stopping	yes
	Patience	5
	Batch size	32
	Epochs	100

3.5 Experimental Result

3.5.1 Performance on Toy datasets

Evaluation Across Machine Learning Models: To assess the adaptability of the proposed representation on discretized data and networks with known structured topologies, we constructed synthetic datasets based on the Barabási–Albert (BA) model, which generates scale-free network structures commonly observed in gene regulatory systems (see Algorithm S2 in the Supplementary Materials). Specifically, 20,000 networks of size 10, and 2,000 networks each of sizes 50 and 100 were generated for training. For validation and testing, the datasets comprised 2,000 networks of size 10 and 400 networks each of sizes 50 and 100. For each ground-truth network, discretized gene expression profiles with three expression levels (0, 1, and 2) were simulated across 50 time points using a stochastic dynamical process. This synthetic setup allows controlled evaluation of the proposed method under well-defined topological and temporal conditions, in

contrast to the inherent variability and noise present in real-world biological data. Detailed statistics of the synthetic datasets are summarized in Table 3.2.

Figure 3.2 compares the AUPR performance of various methods across network sizes 10, 50, and 100, corresponding to 2,000, 400, and 400 test networks, respectively. For each model, we report the mean ± standard deviation of AUPR metric. The baseline random predictor yielded AUPR = 0.268 ± 0.034 (size 10), 0.048 ± 0.003 (size 50), and 0.023 ± 0.001 (size 100), evidencing the growing difficulty of the inference task as network size and hence the candidate edge space increases. Across models, three patterns are salient. First, ensemble/tree-based methods are highly competitive and robust: RF attains the best aggregate performance with AUPR = 0.602 ± 0.093 (size 10), 0.515 ± 0.053 (size 50), and 0.503 ± 0.058 (size 100). LightGBM also performs well at smaller scales (0.413 ± 0.079 at size 10) but degrades more with size. Second, deep learning (MLP) captures complex, non-linear dependencies effectively at small scale (0.566 ± 0.097 at size 10) but its performance declines for larger networks (0.329 ± 0.045 at size 50 and 0.288 ± 0.034 at size 100), consistent with higher sample requirements for stable generalization. Third, GNB exhibits notable robustness at larger scales, achieving AUPR = 0.398 ± 0.141 (size 10), 0.480 ± 0.138 (size 50), and 0.511 ± 0.001 (size 100), which places GNB among the top performers for size 50–100. Other methods (LR, XGBoost, KNN, DT) show variable, often inferior performance depending on scale. These results highlight that the proposed network representation is sufficiently general to support diverse learning paradigms including probabilistic, tree-based, boosting, and neural approaches for the task of structure inference. In addition, model suitability strongly depends on the degree of imbalance and network size, with RF and GNB demonstrating superior robustness under high imbalance, while MLP excels when the network is smaller and data density is higher.

3.5.2 Performance on Escherichia coli datasets

The *E.coli* datasets used in this study were generated using the GNW simulator based on *E. coli* transcriptional regulatory networks. Three network sizes were considered: 10, 50, and 100 genes. For the size-10 configuration, the dataset comprises 200, 20, and 10 network structures for the training, validation, and test sets, respectively. For the larger configurations (sizes 50 and 100), 20 network structures were used for training and 4 for both validation and testing. Each network structure includes 100 distinct time-series experiments, capturing diverse dynamic behaviors under different perturbations. For efficiency and consistency across models, only two representative time-series were selected from each test network during evaluation to ensure comparable computational costs. Detailed dataset configurations and interaction statistics are summarized in Table 3.3.

Consistent with the results observed on the synthetic datasets, the proposed data representation also enables machine learning models to perform effectively on the *E. coli* datasets, the detailed results are visually presented in Figure 3.3. A complete breakdown of the quantitative results for this experimental setting is provided in Supplementary Section 2. Across all network sizes of 10, 50, and 100, the learning-based approaches achieved substantially higher AUPR scores

TABLE 3.2: Summary of the synthetic toy datasets used for GRN inference, including network sizes, number of networks, and interaction statistics

Type	Parameter's name	Network size		
		10	50	100
Training Set	Network structure number	20,000	2,000	2,000
	Time series number per structure	1	1	1
	Time step in each network	50	50	50
	Total interactions (millions)	1.8	4.9	19.8
	Positive versus negative interactions	0.3357	0.0484	0.0292
Validation/Test Set	Network structure number	2,000	400	400
	Time series number per structure	1	1	1
	Time step in each network	50	50	50
	Total interactions (millions)	0.18	0.98	3.96
	Positive versus negative interactions	0.3351/0.334	0.0483/0.0483	0.0234/0.0233

compared with traditional network inference methods such as Jump3, Inferelator, CLR-lag, TSNI, and G1DBN. For instance, at size 10, the average AUPR across all machine learning models reached 0.518 ± 0.251, far exceeding the scores of Jump3 (0.174 ± 0.076) and Inferelator (0.169 ± 0.079). A similar trend was observed for larger networks: at size 50, the average AUPR was 0.164 ± 0.074 compared with 0.050 ± 0.006 for Jump3 and 0.040 ± 0.007 for Inferelator; and at size 100, the corresponding values were 0.111 ± 0.030 versus 0.031 ± 0.005 and 0.027 ± 0.005, respectively. Among the individual models, the GNB consistently achieved the highest and most stable performance across all network sizes 0.566 ± 0.171 (size 10), 0.355 ± 0.083 (size 50), and 0.324 ± 0.055 (size 100). This robustness can be attributed to GNB's probabilistic formulation and its inductive bias of conditional independence, which makes it resilient to high class imbalance and data sparsity known as two persistent challenges in gene regulatory network inference. These results emphasize that the proposed representation enables machine learning models, particularly GNB, to effectively capture the underlying gene–gene dependencies, outperforming traditional inference frameworks and demonstrating scalability to biologically grounded datasets.

We also conducted a series of analyses to assess how the amount of training data and the use of small size networks influence the predictive capability of our learning framework. For this experiment, rather than training all models as in the previous toy dataset evaluation, we specifically selected the GNB model. This choice was motivated by two key reasons: First, GNB demonstrated more stable performance across all network sizes compared to other models, facilitating a clear and consistent comparison. Second, GNB is computationally efficient, allowing faster training and reduced resource usage. Three GNB models were independently trained on size-10 networks, using training sets of 5,000, 10,000, and 20,000 network structures, respectively and evaluated on test sets of sizes 10, 50, and 100. Detailed quantitative outcomes are summarized in Figure 3.4. The complete results for this experimental setting is reported in Supplementary Section 3.

Varying the number of training structures reveals a clear, though saturating, improvement as the training set size increases. With 5,000 structures, the model achieved AUPRs of 0.573, 0.349, and 0.322 on test sizes 10, 50, and 100. Expanding the training data to 10,000 structures resulted in only marginal changes (0.578, 0.352, 0.325), while the 20,000-structure configuration yielded stronger gains primarily for size-10 tests (0.674) and moderate benefits for larger networks (0.373 and 0.333) (see Figure 3.4 **a**). Notably, training on the smallest dataset already provides competitive performance compared with the largest dataset, while requiring substantially less computational time, as illustrated in Figure 3.4 **b** (2.58, 2.86. 3.21). These observations highlight that even limited training sets can suffice to produce near-optimal predictions, offering a favorable trade-off between accuracy and training efficiency.

Importantly, the experiment also highlights a central advantage of our representation: network size becomes far less restrictive, as models trained exclusively on size-10 networks can generalize effectively to larger networks. The size-10 model trained with 20,000 structures attained AUPRs of 0.373 (size 50) and 0.333 (size 100), closely matching models trained directly on size-50 and size-100 datasets (0.359 and 0.326, respectively). In certain cases, the small-network model even outperformed the matching-size baseline. This property is especially valuable when considering the substantial increase in training time: models trained on size-50 and size-100 datasets required 3.79 and 4.27 time units, respectively, which is markedly higher than the time needed for size-10 training. (see Figure 3.4 **b**).

3.5.3 Running time

We further evaluated the computational efficiency of the proposed framework by analyzing its running time in comparison with representative baseline methods. The experiments were conducted on a workstation equipped with an Intel(R) Xeon Silver 4210R CPU at 2.40 GHz, `x86_64` architecture, and 64 GB RAM. All reported values correspond to the logarithm base 10 of the running time measured in milliseconds.

The comparison was performed on three *E. coli* network sizes, namely 10, 50, and 100 genes. For learning-based methods, the running time was decomposed

TABLE 3.3: Summary of the *E.coli* dataset used for GRN inference, including network sizes, number of networks, and interaction statistics

Type	**Parameter's name**	**Network size**		
		10	50	100
Training Set	Network structure number	200	20	20
	Time series number per structure	100	100	100
	Time step in each network	50	50	50
	Total interactions (millions)	1.8	4.9	19.8
	Positive versus negative interactions	0.1	0.08	0.01
Validation/ Test Set	Network structure number	20/10	4	4
	Time series number per structure	100/2	100/2	100/2
	Time step in each network	50	50	50
	Total interactions (millions)	$0.18/1.8 \times 10^{-3}$	0.98/0.0196	3.96/0.079
	Positive versus negative interactions	0.18/0.156	0.041/0.044	0.025/0.023

into two components: training time and inference (testing) time. In contrast, methods such as Jump3, Inferelator, CLR-lag, TSNI, and G1DBN do not involve an explicit training phase; therefore, their reported running time corresponds to the end-to-end inference time, measured from the start of execution until the inferred network is obtained. Details of the datasets used for training and testing are provided in Table 3.3.

As shown in the Figure 3.5, the running time comparison across all three network sizes, the training time of learning-based methods is generally higher than the inference time of direct GRN inference algorithms such as Jump3 and CLR-lag. This behavior is expected, as the former require an explicit model learning phase, whereas the latter directly infer the network structure from the input data without prior training. However, once trained, the learning-based methods exhibit substantially lower inference time compared to the end-to-end inference time of the direct methods.

This distinction becomes particularly important in scenarios where a large number of networks need to be inferred. For the proposed framework and other learning-based approaches, the training process is performed only once, after

which the trained model can be reused to infer multiple networks efficiently. Consequently, the overall computational cost scales primarily with the inference time, which remains relatively small. In contrast, direct inference methods incur a similar computational cost for each network, as the entire inference procedure must be executed independently for every instance. As a result, their total running time increases linearly with the number of networks, making them less efficient when applied to large-scale or repeated inference tasks.

3.6 Discussion and Conclusion

The proposed framework demonstrates stable performance across both discrete and continuous datasets, highlighting robustness under diverse data generation regimes in gene regulatory network inference. By introducing an interaction level representation that jointly encodes temporal expression dynamics and regulatory structure, the framework enables the effective use of a broad range of machine learning classifiers and consistently outperforms traditional approaches based solely on temporal profiles. A key advantage of the framework is its assumption free design. In contrast to many existing methods that rely on heuristic choices such as predefined time lags or fixed candidate regulator sets, the proposed representation avoids explicit parametric assumptions about regulatory dynamics. This improves robustness across datasets and reduces the need for extensive parameter tuning. From a practical standpoint, the framework provides a well-defined inference protocol for unseen data. Inference is feasible when the temporal length of new expression profiles matches or exceeds that used during training through temporal alignment. When time series are shorter, direct inference is not possible and temporal extension techniques are required (Anh and Y.-K. Kwon, 2025), explicitly defining the current scope and limitation of the method. Importantly, the proposed representation is scale invariant, as regulatory interactions are encoded independently of global network size. This property enables effective generalization across networks of different sizes and allows training data to be augmented with interaction samples from heterogeneous network scales, mitigating class imbalance. Together, by addressing shared limitations of existing task transformation approaches, including fragile assumptions, limited scalability, and weak cross network generalization, the proposed framework provides a flexible and scalable foundation for time series GRN inference and supports robust deployment in practical biological settings.

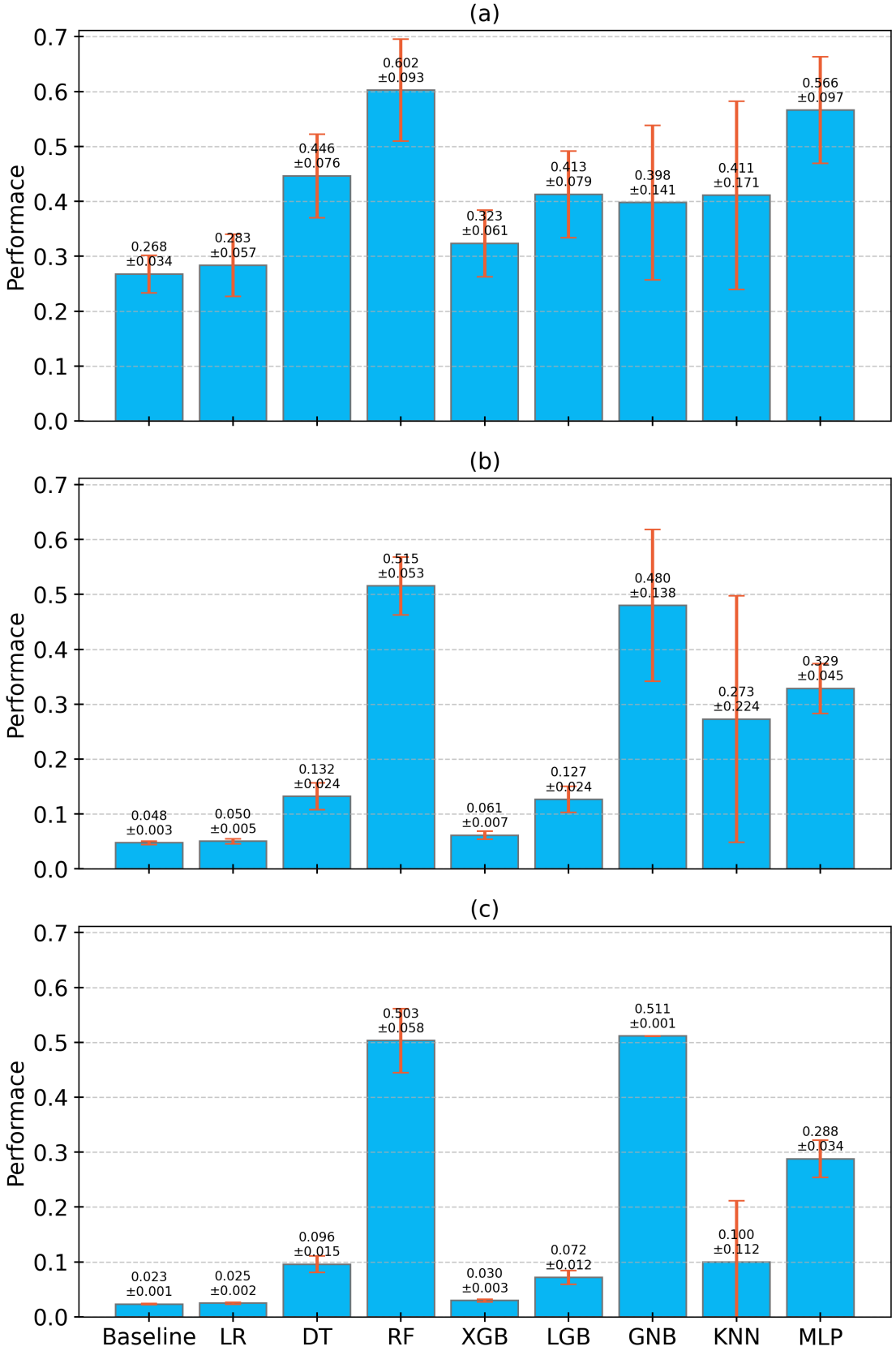


FIGURE 3.2: Figure shows the AUPR performance of different methods along the X-axis on the toy datasets. The Y-axis indicates the average AUPR values, with error bars representing the standard deviation. Subplots (**a**), (**b**), and (**c**) correspond to network sizes 10, 50, and 100, respectively, evaluated on 2,000, 400, and 400 test networks.

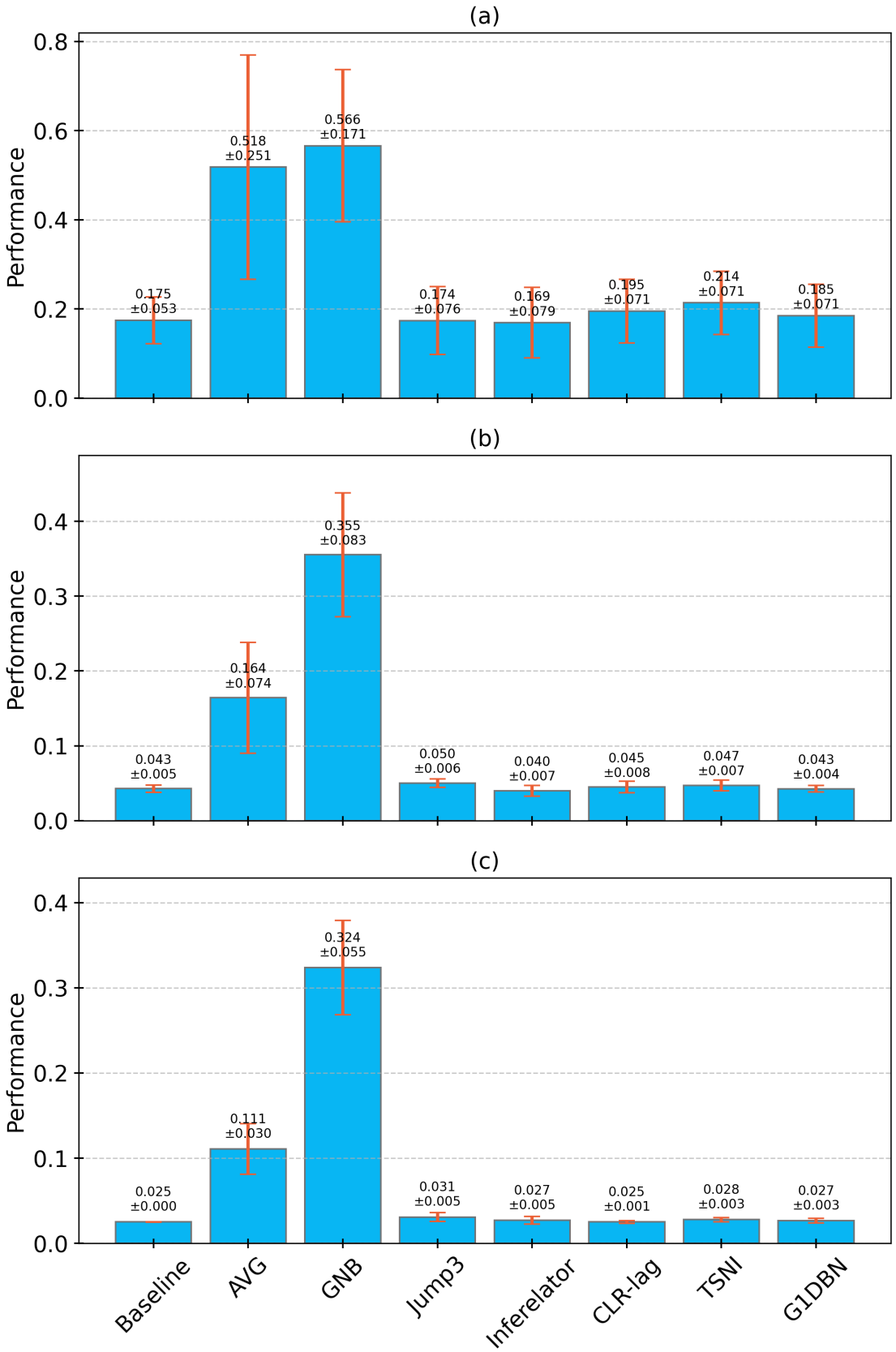


FIGURE 3.3: Figure shows the AUPR performance of different methods along the X-axis on the *E.coli* datasets. The Y-axis indicates the average AUPR values, with error bars representing the standard deviation. Subplots (**a**), (**b**), and (**c**) correspond to network sizes 10, 50, and 100, respectively, evaluated on 20, 8, and 8 test networks.

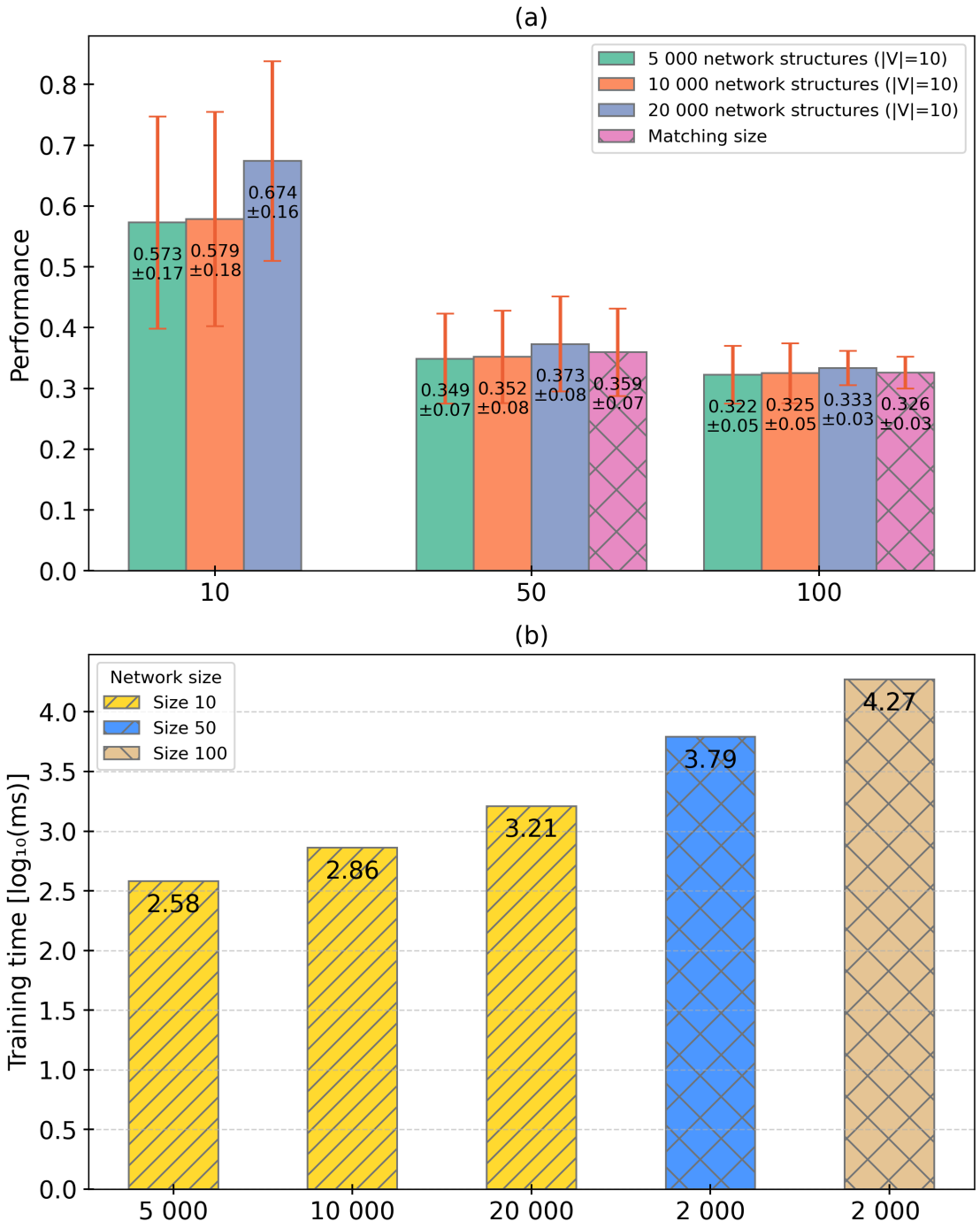


FIGURE 3.4: Performance and training-time analysis across varying training-set sizes and network scales. (**a**) Comparison of AUPR performance for three GNB models trained on datasets containing 5,000, 10,000, and 20,000 network structures (all with training network size of 10). Each model is evaluated on test networks of sizes 10, 50, and 100. Hatched bars denote the performance of models trained on matching network sizes (i.e., size-50 and size-100 training sets tested on size-50 and size-100 networks), included as baselines for comparison. The x-axis indicates the size of the testing network, and the y-axis reports AUPR. (**b**) Training-time comparison across datasets differing in both the number of network structures and the network size. The x-axis lists the training-set configurations (size-10 with 5 000, 10 000, and 20 000 structures, followed by size-50 and size-100 datasets, each has 2 000 network structures), and the y-axis reports the training time in log10-scaled milliseconds.

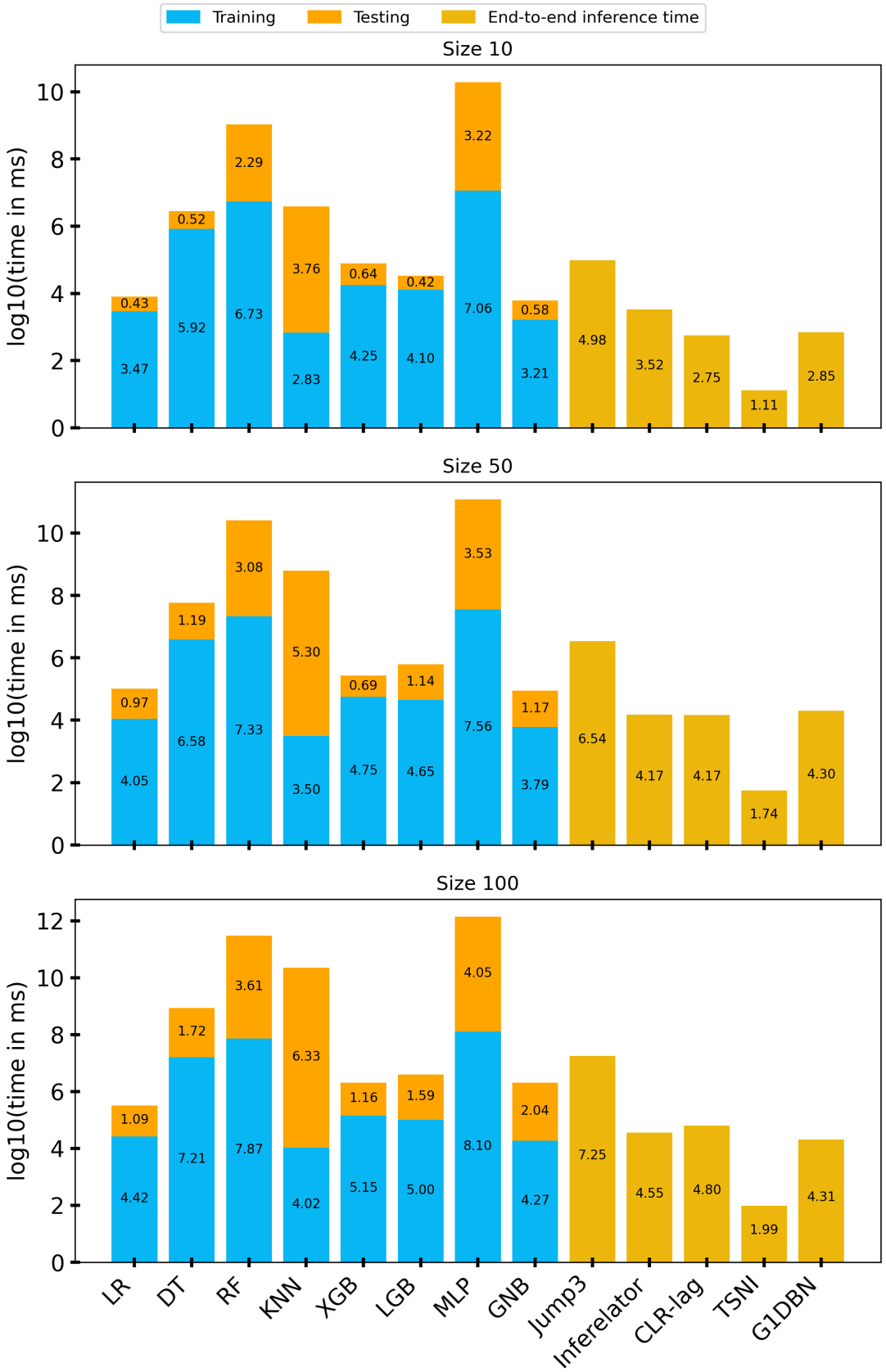


FIGURE 3.5: Comparison of running time across different GRN inference methods on the *E. coli* dataset for network sizes 10, 50, and 100. The y-axis shows the logarithm base 10 of the running time measured in milliseconds, while the x-axis represents the evaluated inference methods. Results highlight the trade off between higher initial training cost and faster inference for learning-based methods, particularly in scenarios involving repeated network inference.

Chapter 4

Synthesizing the Time-series Gene Expression Data to Enhance the Network Inference Performance Using the Autoencoder

4.1 Abstract

It is a challenge to infer a gene regulatory network from the time-series gene expression data in the systems biology field. Lack of gene expression data samples is a factor limiting the performance of the inference methods. To resolve this problem, we propose a novel autoencoder-based approach that synthesizes virtual gene expression data to be used as input to the inference method. Through intensive experiments, we showed that using the synthetic gene expression as input improves the performance of the network inference method compared to that without it. In particular, the performance improvement was stable against the discretization level of gene expression, the number of time steps in the observed gene expression, and the number of genes.

4.2 Introduction

GRNs describe the collective connections among genes and regulate their expressions. It is essential to clarify GRNs to understand the function and disease of organisms. Thus, the inference of GRNs has been an important research topic (Malvina and Pancaldi, 2023), and many diverse methodologies which are usually classified into model-free and model-based methods have been developed. In model-free approaches, gene interactions can be identified by various statistical and machine learning techniques including mutual information (Margolin et al., 2006; Yang et al., 2018; Liang and X. Wang, 2008), random forests (**HuynhThu2010**; Petralia et al., 2015; Park et al., 2018), or network deconvolution (H. Chen et al., 2014; Siegal-Gaskins, Ash, and Crosson, 2009). In model-based methodologies, a quantitative dynamical model is employed to represent the system's dynamics such as the Boolean networks (Trinh and Y. Kwon, 2021; Hickman and Hodgman, 2009), ordinary differential equation (ODE)(Ma, Fang, and Jiao, 2020; Z. Li et al., 2011; Aalto et al., 2020; Bin and Y. Chen, 2020), regression method (Thorne, 2018; Michailidis and d'Alché-Buc, 2013; Breiman, 2001),

or dynamic Bayesian networks (DBN) (K. Murphy and Mian, 1999; Kim, Imoto, and Miyano, 2003; Zou and Conzen, 2005).

In the model-based inference methods, it is necessary to optimize the model parameters by using gene expression data. It is obvious that the data from longer observation times would help infer GRNs more accurately. However, time series gene expression data is not always easily obtained (D. Murphy, 2002). In some cases, it may be difficult to ensure that observations are made under the identical conditions throughout the observation period. Data augmentation has been considered as one of the effective approaches to resolve this data shortage problem. It has been frequently used in various domains such as image processing (Waqas et al., 2022; Frid-Adar et al., 2018; Y. Zhang, Q. Wang, and B. Hu, 2023; Shin et al., 2018) and time series prediction (Forestier et al., 2017; Maweu et al., 2021; Leznik et al., 2021). In the field of time series gene expression, several strategies have been also proposed to generate artificial data, in particular, through the Dialogue on Reverse Engineering Assessment and Methods (DREAM) challenge (Schaffter, Marbach, and Floreano, 2011; DREAM, 2009). Several previous investigations on this matter are accessible (I. Albert et al., 2008; H. d. Jong, 2002; H. D. Jong et al., 2003; Mendes, Sha, and Ye, 2003; Bulcke et al., 2006). These studies can generate simulated biological datasets that closely mimic real-world data, and provide researchers with tools to analyze and investigate genes. The Artificial Gene Networks (AGN) model is proposed to simulate temporal expression data for the identification of gene networks (Lopes, Jr., and Fontaura Costa, 2011). This model uses a feature selection technique where a specific gene remains constant, and the expression profiles of all other genes are observed. The goal is to identify a significant subset of predictors for the target gene. These studies proposed methods to produce synthetic data that closely resemble the real data. However, they have a limitation in producing a variety of data because they did not learn complex generation functions using data but tried to build simple rules. In particular, rule-based or manually designed simulations may fail to capture nonlinear or context-specific regulatory dynamics that naturally occur in biological systems. Additionally, these approaches often lack adaptability and may not generalize diverse datasets or unseen biological conditions.

In this study, we propose a network inference approach involving an autoencoder-based data generation model. For our problem, it is noteworthy that the autoencoder has some advantages over the generative adversarial network (GAN) (Goodfellow et al., 2020). The autoencoder provides a simple yet effective framework with a lightweight architecture that enables stable training by optimizing a single objective. On the other hand, the GAN requires the simultaneous training of two competing networks which causes mode collapse, sensitivity to hyperparameters, and instability during training (Arjovsky and Bottou, 2017; Salimans et al., 2016). Thus, the autoencoder is more suitable for gene expression problems, where available datasets are typically small and heterogeneous. Several studies have shown that autoencoders perform well in low-data regimes and can effectively extract meaningful representations in biological applications (Tan et al., 2015; L. Chen et al., 2016). In addition, the performance of autoencoders is

less dependent on the underlying distribution of gene expression values, allowing better adaptability to biological variation. In this context, we utilize an autoencoder to learn a functional representation of gene expression dynamics from limited time-series data, generating additional synthetic samples to support more accurate network inference.

4.3 Materials and Methods

4.3.1 Discretized Network Model

In this study, we used a gene-wise discretized network model as presented in the previous study (Anh and Y.-K. Kwon, 2023). The discretized network can be modeled as a directed graph $G(V, A)$, where the vertex set $V = \{v_1, v_2, \ldots, v_N\}$ corresponds to N genes, and edges $A \subseteq V \times V$ encode regulatory interactions between them. The state of a gene v at time t, denoted as $v(t)$, assumes one of l discrete values from the set $\{0, 1, 2, \ldots, l-1\}$. When $l = 2$ for all genes in V, the network is referred to as a Boolean network (Kauffman, 1971). Suppose that $v \in V$ denotes a target node whose dynamics are governed by the interaction with k regulatory genes $u_1, u_2, \ldots, u_k$, where $u_i \in V$. Let E and E_i denote the sets of possible discrete expression values for the gene v and the regulatory genes u_i, respectively. The state of v at the next time step, $v(t+1)$, is determined by a discrete function $f : E_1 \times E_2 \times \cdots \times E_k \to E$, which maps the expression values of the k regulatory genes at time t to an updated state. Consequently, the update rule for v can be expressed as follows:

$$v(t+1) = f(u_1(t), u_2(t), ..., u_k(t)). \tag{1}$$

In this study, we adopted an update time lag of one. The total number of possible functions f is given by $l^{\prod_i l_i}$, where $l = |E|$ and $l_i = |E_i|$ denote the cardinalities of the sets E and E_i, respectively.

4.3.2 Network Inference Problem

In this study, we addressed the problem of network inference from time-series gene expression data, aiming to recover both the regulatory interactions and their associated update functions. The quality of the inferred network was evaluated by comparing the trajectories it generates with those observed in the original time-series data. Let $\mathbf{v}' = [v'(1), v'(2), ..., v'(T)]$ denote the sequence of the predicted expression states of gene v for a time step ranging from 1 to T, based on the inferred discretized network. Then, the dynamic accuracy of the inferred network is defined as follows:

$$DynamicAccuracy = \frac{1}{N \cdot T} \sum_{i=1}^{N} \sum_{t=1}^{T} (1 - D(\mathbf{v}', \mathbf{v})), \tag{2}$$

where T denotes the total number of time points and $D(\mathbf{v}', \mathbf{v})$ is the Hamming distance between two sequences of $\mathbf{v}'$ and $\mathbf{v}$.

4.3.3 Structural Performance Metrics

The performance of the proposed method was evaluated using the same metrics as those described in Section 2.3, namely precision, recall, and structural accuracy. These metrics were used to assess the structural similarity between the inferred networks and their corresponding ground-truth networks. For consistency, their definitions and calculation procedures remain unchanged from those presented in the previous chapter.

4.3.4 Network Inference Method

In this study, we synthesized time-series discrete expression data to improve the existing network inference method. To this end, we used mutual information based on multiple-level discretization network inference (MIDNI) (Anh and Y.-K. Kwon, 2023). The latter is an algorithm developed from the mutual information-based Boolean network inference method (MIBNI) (Barman and Y. K. Kwon, 2017). The former reconstructs the discrete network by properly discretizing gene expression values into two or three levels depending on their distribution, whereas the latter infers the Boolean network based on the binarized gene expression values. This higher degree of freedom in expression led to more accurate inference results in MIDNI.

4.3.5 Autoencoder Model

The concept of an autoencoder was originally proposed by LeCun in 1987 (LeCun, 1987). It is an artificial neural network designed to acquire efficient representations of input data, primarily for dimensionality reduction or feature extraction. It comprises two components, namely an encoder and a decoder. The encoder compresses the input data into a reduced-dimensional representation, referred to as the "latent space" or "code", whereas the decoder utilizes the compressed latent representation to recreate the original input data with maximal fidelity. Since it was invented, numerous autoencoder variations have been effectively used in a variety of domains, including natural language processing (J. Li, Luong, and Jurafsky, 2015; Freitag and Roy, 2018; Akram et al., 2022), computer vision (Zhou and Liu, 2023; Parmar et al., 2021; Cheng et al., 2018), and speech recognition (Vachhani et al., 2017; Karita et al., 2019). In our study, we employed a conventional autoencoder with a reasonable number of hidden layers to achieve a compromise between efficiency and training duration (Géron, 2022; Song, Jin, and X. Hu, 2022). Unlike traditional autoencoders that aim to reconstruct the input data itself, the proposed model is trained to predict the next state $v(t+1)$ given the current state $v(t)$ as input. This architecture retains the encoder–decoder structure, but the training objective shifts from input reconstruction to temporal state prediction. It seeks to reduce the discrepancy between the actual and predicted values by minimizing the loss function. By learning a compressed latent representation of $v(t)$ that captures temporal dynamics, the decoder is optimized to generate the corresponding $v(t+1)$ rather than reproducing $v(t)$. While the model may produce diverse gene expression profiles due

to the limited dataset, this variation is consistent with natural biological variability. Instead of evaluating the biological plausibility of the synthetic data directly, we assess its utility through downstream network inference, interpreting the successful reconstruction of gene regulatory networks as an indirect yet meaningful indicator of data quality.

4.4 Our Proposed Method

Figure 4.1 shows the entire framework of our approach. An observed discrete gene expression dataset is given as an input. It is represented by a $K \times N$ discrete matrix, where K and N indicate the number of observed time steps and genes, respectively. The matrix is used to generate two other discrete matrices of $(K-1) \times N$ size, which are input–target pair matrices. They establish a relationship between consecutive network states, which will be learned by the autoencoder. The learned autoencoder can synthesize gene expression values after the observed time steps. The synthesized and observed discretized gene expressions are combined into a single matrix, and this is used as input for the inference algorithm MIDNI. Each part of our approach is explained in detail in the next subsections.

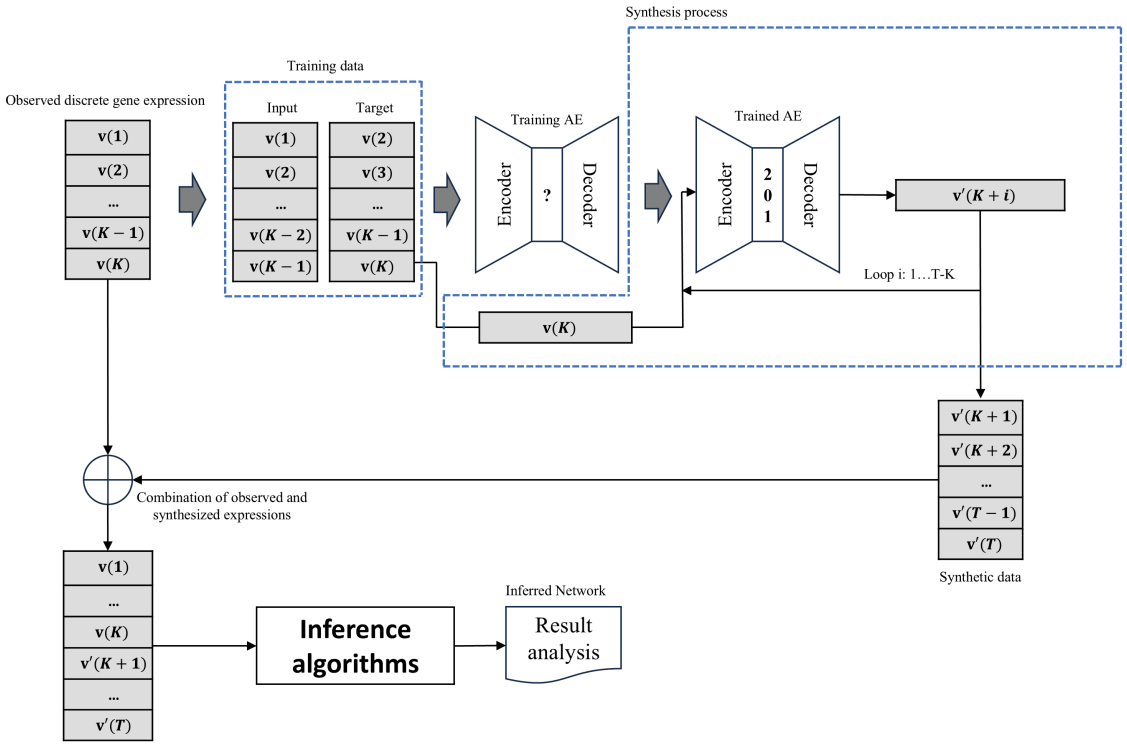


FIGURE 4.1: The framework of our method. The observed expression data create the input–target matrices, which represent a pair of consecutive expression vectors to train an autoencoder. The trained autoencoder autoregressively synthesizes expression data. The combined data of the observed and synthesized expressions are used as input for the inference algorithms.

4.4.1 Synthesis of Gene Expression

The given input data is an observed, discretized gene expression matrix of $K \times N$ size. Let $\mathbf{v}(t) = [v_1(t), v_2(t), ..., v_N(t)]$ be a network state at time step t represented by a vector of all gene values. Then, the input data are the collection of $\mathbf{v}(t)$ for all observed time steps ($t = 1, 2, \ldots, K$). The observed gene expression matrix is transformed into two $(K-1) \times N$ matrices called the input and target matrices for training the autoencoder. The first and second matrices are the collections of $\mathbf{v}(t)$ ranging from $t = 1$ to $t = K-1$ and from $t = 2$ to $t = K$, respectively. This representation can delineate the correlation of gene expression values between two consecutive time steps, enabling the autoencoder to effectively identify the underlying dynamics. After the training process is finished, the autoencoder model can be used to generate unobserved discrete gene expression data after the last observation time step K. At the first iteration of the generation process, $\mathbf{v}(K)$, the network state at the last observation time step is given as an input to the model, and the model produces an output $\mathbf{v}'(K+1)$, which is used as an input for the output $\mathbf{v}'(K+2)$ and so on. This process is repeated for a desired time step. The structure of the autoencoder is presented in Table 4.1. We chose the set of parameters that provides the best results, as presented in Table 4.2. The model structure and hyperparameters were selected through extensive empirical tuning considering the specific characteristics of biological gene expression data and the demands of network reconstruction tasks. A symmetric three-layer architecture was chosen to provide sufficient capacity for capturing the complex nonlinear relationships commonly observed in gene regulatory networks while avoiding overfitting typically observed in small biological datasets. ReLU activation functions were used in the hidden layers due to their ability to efficiently learn sparse representations. This aligns with the biological assumption that only a subset of genes is co-regulated at a time. In contrast, alternatives like sigmoid and tanh introduced saturation effects and slower convergence. Linear output activation preserves the full dynamic range of gene expression values (between 0 and 2). The learning rate and training epochs were dynamically adjusted to accommodate datasets of varying size and complexity, ensuring stable convergence without sacrificing generalization. These design choices were made to optimize the model's ability to generate biologically plausible data and support the accurate inference of regulatory interactions.

TABLE 4.1: The structure of the autoencoder model.

Parameter's Name	**Value**
Dimension of the input layer	N
Dimension of the output layer	N
Number of hidden layers	3
Number of neurons in the first hidden layer	$2 \times N$
Number of neurons in the second hidden layer	512
Number of neurons in the third hidden layer	$2 \times N$
Activation function in hidden layers	relu
Activation function in the output layer	linear

TABLE 4.2: Parameters of the training.

Parameter's Name	Value
Model optimizer	Adam
Learning rate	0.001–0.01 *
Loss function	mse
Number of epochs	1000–10,000 *
Batch size	32

* The values are different depending on the size of the input network.

4.4.2 Inference of GRN

As mentioned in Subsection 4.3.4, we applied MIDNI to the synthesized discrete gene expression data. Two key parameters are configured for MIDNI in this study: First, the time lag parameter is set to one, that is to say, the expression of a regulator gene at time step t influences the expression of its target gene at time step $(t+1)$. The time lag accounts for the inherent delays in biological processes, such as transcription, translation, and the diffusion of regulatory molecules, which are essential for accurately capturing the dynamic nature of gene regulatory interactions. Second, the maximum number of regulators is set to five. This constraint reduces the computational complexity of the network inference process while maintaining biological relevance such that most genes are regulated by a relatively small number of key regulators in real biological systems.

4.5 Experimental Results

4.5.1 Experiment Setup

To comprehensively evaluate the performance and robustness of our proposed method, we conducted experiments on synthetic datasets representing GRNs of two different network sizes $N = 50$ and $N = 100$. These sizes were chosen to reflect small-to-intermediate complex biological systems, allowing us to examine the scalability of our method under different conditions. For each network size, we generated 10 structurally distinct ground truth networks using the Barabási–Albert (BA) model (Barabási and R. Albert, 2006), which is a widely used generative model that captures the scale-free properties commonly observed in real-world biological networks. This resulted in a total of 20 unique ground truth networks used in our experiments. For each ground truth network, we simulated discretized gene expression data over K observed time steps using a dynamic model outlined in Algorithms 1 and 2 (see Appendix A). These simulated time-series datasets serve as the foundation for evaluating our method's ability to infer network structure from temporal gene expression patterns. Two levels of discretization were considered to reflect different degrees of resolution in experimental measurements, namely binary (two states: 0 or 1) and ternary

(three states: 0, 1, or 2). This allowed us to assess the method's sensitivity and effectiveness across datasets with varying levels of granularity. Our method takes the observed gene expression data over the initial K time steps as input and synthesizes the gene expression values for the remaining $(T - K)$ time steps, where the total length of the time-series is fixed at $T = 100$. The ability to generate synthetic future data is a critical component of our approach, as it allows us to enrich the dataset and potentially improve inference accuracy, especially in cases where only a limited number of observations are available. To systematically investigate the influence of the number of observed time steps on inference performance, we varied K from 10 to 90 in increments of 10. This setup enables us to explore how the quantity of available temporal data affects both the quality of synthetic data generation and the subsequent network inference. We applied the network inference algorithm, MIDNI, to both the original and synthesized datasets. The inferred networks were then compared against the corresponding ground truth networks using the evaluation metrics explained in Section 4.3.3. Performance was analyzed separately for datasets discretized into ternary levels (see Figure 4.2) and binary levels (see Figure 4.3). Comprehensive results are provided in Tables B.1 and B.2 (see Appendix B). These comparisons provide insight into how well our method reconstructs the underlying network structure under different data conditions and discretization schemes.

4.5.2 Results of Structural Performance

As shown in Figure 4.2a (see Table B.1 in Appendix B for details), the precision values of the networks inferred based on the original data (dashed line) and the synthetic data (solid line) continuously increase as the observed time steps (K) increase in the networks with $|V| = 50$. This is because the more expression data used for the inference algorithm, the higher the inference performance. The precision improvement by the use of the synthetic data was observed in all K values except for $K = 60$ and 90. In addition, we can observe a similar trend with respect to the recall performance (Figure 4.2b). In particular, the degree of the performance improvement at $K = 10$ was significantly larger. These observations indicate that the precision improvement by the synthetic data is not by reducing the number of positive predictions of interactions but by predicting the true positive interactions more accurately. Although the synthetic data also improved the structural accuracy, the improvement was not significant (Figure 4.2c). This is because the number of true negatives is much larger than that of true positives. We note that these findings were consistently observed in the networks with a larger network size ($|V| = 100$) (Figure 4.2d–f). Taken together, our synthetic approach can be considerably useful when the observation time of the gene expression is limited in the real experimental environment.

We also investigated our method with respect to the Boolean networks, as shown in Figure 4.3 (see Table B.2 in Appendix B for details). As in the ternary network results, we can observe the performance improvement of precision, recall, and structural accuracy in most cases. This means that our method is also effective in the Boolean network models. It is interesting that the recall improvement by our method was relatively larger on average than the precision improvement in the case of $|V| = 50$, whereas the latter was relatively larger than the first

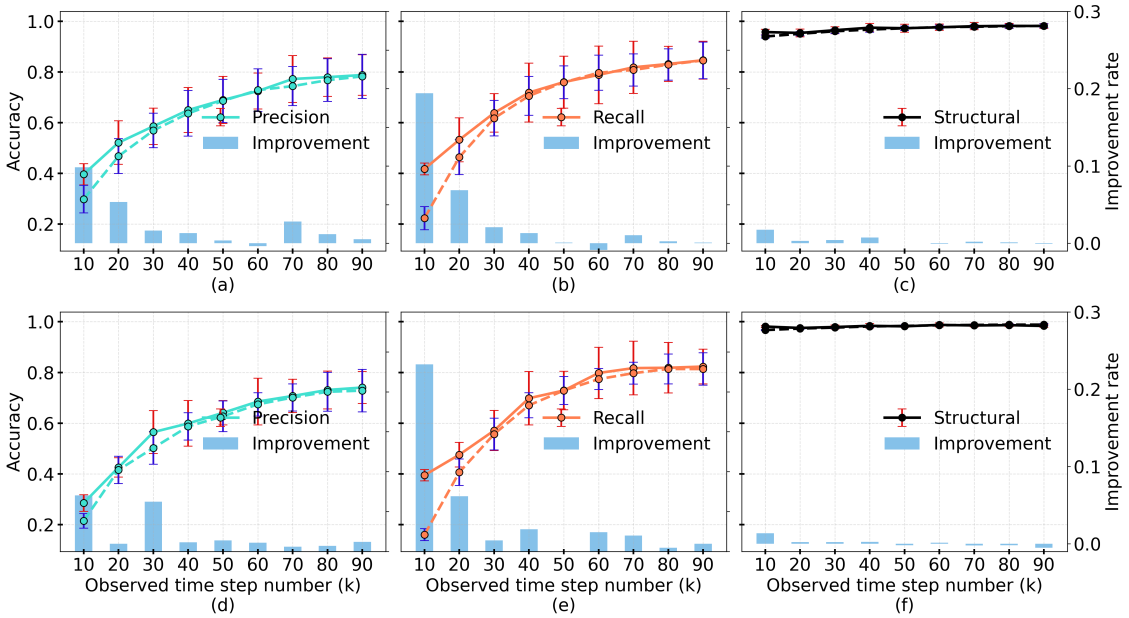


FIGURE 4.2: Results of precision, recall, and structural accuracy on the multi-level discretization dataset. (**a**–**c**). Results for the network with $|V| = 50$. (**d**–**f**).Results for the network with $|V| = 100$. The Y-axis denotes the performance metrics, and the X-axis indicates the number of observed time steps in the original gene expression data. For the synthesized dataset, gene expression data are generated for $(100 - K)$ time steps. Solid and dashed lines represent the results based on the synthesized and the original dataset, respectively. Each point and error bar indicates the average value and the standard deviation, respectively, over 10 random networks. The bar charts show the performance improvement of the synthesized dataset over the original dataset.

in the case of $|V| = 100$. This is because the MIDNI algorithm cannot control the number of positive predictions. However, we note that there is no negative influence on any performance measure. It is also interesting that the performance improvement does not increase as the observation time step K increases. For example, the precision was highest when K is 60 and 50 in the networks of $|V| = 50$ and $|V| = 100$, respectively (Figure 4.3a,d), and the recall was highest at $K = 70$ in the networks of $|V| = 100$ (Figure 4.3e). These results mean that the desirable number of observation time steps is dependent on the degree of discretization levels.

4.5.3 Results of Dynamic Accuracy

We also evaluated the dynamic accuracy of both the networks inferred from the original and the synthetic data on the multi-level discretization dataset (Figure 4.4) and on the Boolean discretization dataset (Figure 4.5). Contrary to precision or recall metrics, the dynamic accuracy tends to decrease as the number of time steps increases. We believe this is due to the increase in the number of data

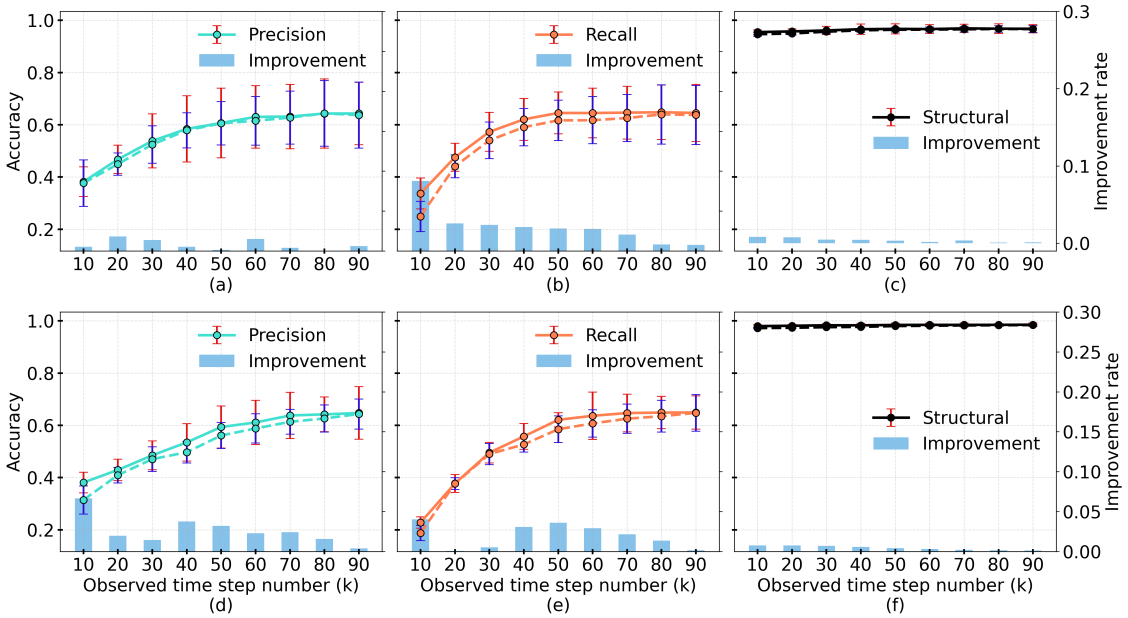


FIGURE 4.3: Results of precision, recall, and structural accuracy on the Boolean discretization dataset. (**a**–**c**) Results for the network with $|V| = 50$. (**d**–**f**) Results for the network with $|V| = 100$. The Y-axis denotes the performance metrics, and the X-axis indicates the number of observed time steps in the original gene expression data. For the synthesized dataset, gene expression data are generated for $(100 - K)$ time steps. Solid and dashed lines represent the results based on the synthesized and the original dataset, respectively. Each point and error bar indicates the average value and the standard deviation, respectively, over 10 random networks. The bar charts show the performance improvement of the synthesized dataset over the original dataset.

samples to be fitted by the inferred network. Moreover, the incorporation of additional time points may introduce deviations from the patterns already learned by the model, particularly if the new data include noise, nonlinear dynamics, or previously unobserved system behaviors. In addition, the dynamic accuracy of the synthetic data-driven networks is inferior to that of the original data-driven networks in most cases. We note that the latter is investigated over K time steps, whereas the former is investigated over T time steps, which is longer than K. Considering this difference between the examination time steps, the performance difference in terms of the dynamics accuracy may not be considerably significant.

4.5.4 Results of Similarity of Synthesized Expressions

The autoencoder can generate the gene expression data differently whenever it is executed. To investigate how much the synthesized data are varied, we first created a random ground truth network with $N = 100$ and generated an original ternary gene expression with K time steps ranging from 10 to 90 (see Algorithm 2 in Appendix A). For each K, we trained an autoencoder using the original

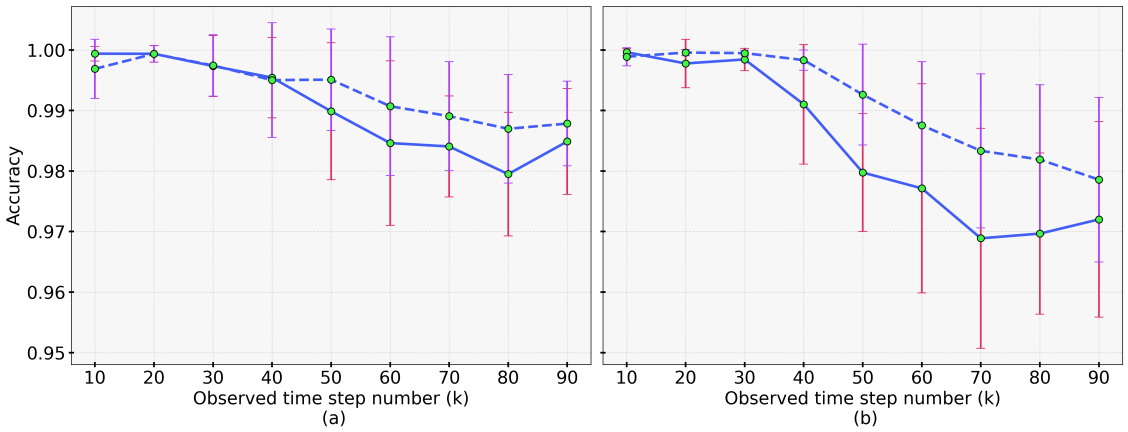


FIGURE 4.4: Comparison results of dynamic accuracy on the multi-level discretization dataset. (**a**) Networks of $|V| = 50$. (**b**) Networks of $|V| = 100$. The Y-axis denotes the dynamic accuracy, and the X-axis indicates the number of observed time steps. The dashed and solid lines represent the results of the networks inferred from the original and the synthetic expression data, respectively.

gene expression and generated three different synthetic gene expressions for later $= T - K$ time steps. For three pairs of the gene expression matrices of $T \times N$ size, we investigated the similarity matrix, where the white and the black dots represent whether the corresponding gene expression values in the pair matrices are identical or not, respectively (Figure 4.6). As shown in the figure, the synthesized gene expressions are significantly different from each other. However, the inference performance based on these synthetic data is not significantly different. In other words, the autoencoder model produces diverse synthesized expressions but induces stable performance.

4.6 Discussion and Conclusions

In this study, we used an autoencoder model to synthesize gene expression data to overcome data scarcity problems in reverse engineering. Specifically, it produces additional time-series expression data to complement the observed gene expression data. The synthesized gene expression data were assessed using an inference algorithm. The experimental results demonstrated that the inference performance using synthetic gene expression was consistently higher than that using only the observed gene expression. This finding substantiates the effectiveness of the proposed model as a valuable tool for addressing network inference problems, particularly when there is a limitation in obtaining gene expression data of a sufficiently long observation time.

Furthermore, as gene expression data inherently possess structural relationships among genes, future work could explore the integration of graph representation learning techniques as a complementary or alternative approach. These

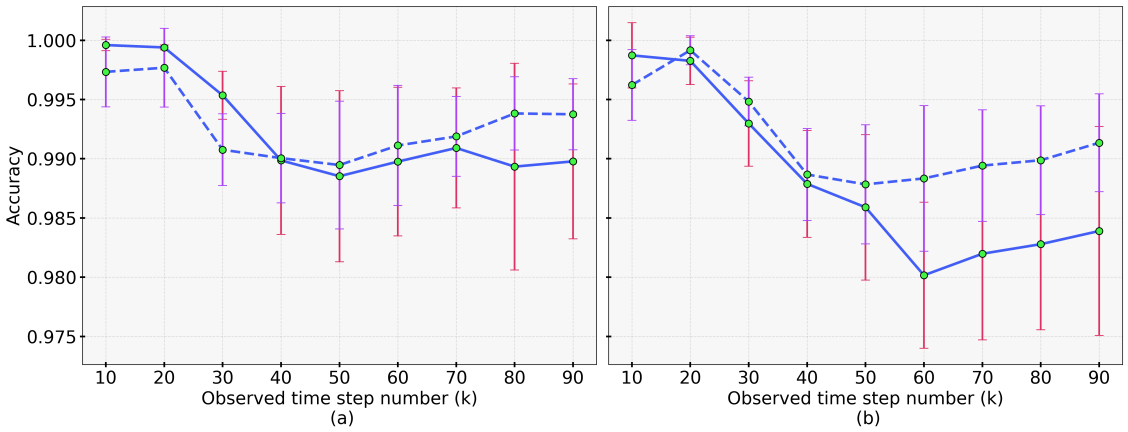


FIGURE 4.5: Comparison results of the dynamic accuracy on the Boolean discretization dataset. (**a**) Networks of $|V| = 50$. (**b**) Networks of $|V| = 100$. The Y-axis denotes the dynamic accuracy, and the X-axis indicates the number of observed time steps. The dashed and solid lines represent the results of the networks inferred from the original and the synthetic expression data, respectively.

methods are well-suited for modeling structured biological systems and may improve the quality of synthetic data or enhance network inference performance by capturing complex topological patterns in gene regulatory networks (F. Chen et al., 2020; G. Li et al., 2024). Another future study could be to extend the current framework based primarily on discretized gene expression data to operate directly on continuous-valued datasets. The reliance on discretization introduces variability due to method-specific assumptions, potentially affecting inference outcomes. Developing a discretization-free approach would increase the generalizability of the method. Additionally, integrating more advanced generative models such as GANs may further improve the diversity and biological realism of the synthesized data. Finally, a comprehensive validation of real-world biological datasets will be essential to evaluate the robustness and practical utility of the proposed method across diverse application scenarios, including rare disease modeling and temporal transcriptomic analyses.

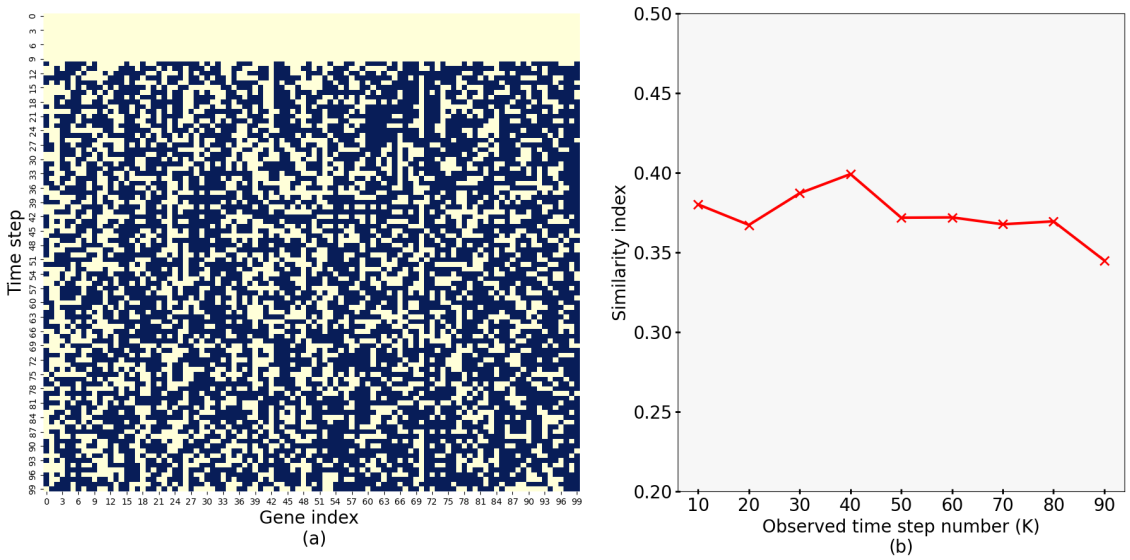


FIGURE 4.6: Similarity matrix between the synthesized gene expression data. (**a**) An example of a similarity matrix between a pair of synthesized gene expression data values when K equals 10. The white or black dot indicates that the corresponding gene values between two expression data values are identical or not, respectively. (**b**) Change in the mean similarity index against the time step number (K). For each K from 10 to 90, three pairs of synthesized gene expression data values were used to compute the mean similarity index. The Y-axis denotes the similarity index, defined as the rate of the number of identical dots over the total number of dots in the gene expression matrix. The X-axis corresponds to the number of observed time steps K.

Chapter 5

General Discussion, Limitations, and Future Directions

5.1 General Overview of the Thesis

The central objective of this thesis is to address the long-standing challenge of inferring GRNs from time series gene expression data. This problem is fundamentally difficult due to the high dimensionality of biological systems, the limited number of observed time points, the presence of noise, and the intrinsic multimodality of network dynamics. Rather than focusing solely on improving inference algorithms, this thesis adopts a broader perspective, emphasizing the critical roles of data representation and data availability in enhancing GRN inference performance.

Chapter 2 introduces a multiple-level discretization framework for network inference. Unlike most existing discretization-based methods that rely on a fixed Boolean representation using two discrete states, this chapter proposes a flexible discretization strategy in which each gene can be represented using either two or three discrete levels, depending on the distribution of its expression values. By avoiding a rigid discretization scheme and allowing gene-specific discretization levels, the proposed approach preserves more biologically meaningful information while maintaining computational efficiency. This adaptive representation significantly improves both inference accuracy and runtime performance, demonstrating that discretization, when properly designed, can be a powerful tool rather than a limiting factor in GRN inference.

Building upon the importance of data representation, Chapter 3 proposes a novel representation learning framework that jointly encodes gene expression dynamics and network structural information. Instead of directly inferring regulatory relationships from raw or discretized expression profiles, this framework transforms the GRN inference problem into a supervised learning task. By constructing informative feature representations that capture both temporal expression patterns and structural dependencies, standard machine learning classifiers can be effectively applied for network prediction. This chapter highlights that learning suitable representations can bridge the gap between biological network inference and modern machine learning techniques, enabling more scalable and flexible inference models.

While Chapters 2 and 3 focus on improving inference performance through

better representations, Chapter 4 addresses a complementary but equally important challenge: data scarcity. In many practical scenarios, the number of observed time points is severely limited, which substantially degrades inference accuracy. To mitigate this issue, Chapter 4 proposes a data synthesis approach that generates additional time-series samples using an autoencoder-based model. By augmenting the original dataset with synthesized expression trajectories, the proposed method enhances the robustness and reliability of network inference under low-data conditions. This chapter demonstrates that improving data quantity, when done carefully, can significantly boost inference performance without requiring additional experimental measurements.

Taken together, the contributions of this thesis demonstrate that accurate GRN inference is not solely determined by the choice of inference algorithm. Instead, it critically depends on how gene expression data are represented, how structural information is incorporated, and how data limitations are addressed. By systematically exploring these aspects across Chapters 2 to 4, this thesis provides a comprehensive framework for improving GRN inference from time-series gene expression data.

5.2 Cross-Method Discussion and Integration of Findings

5.2.1 Data Representation as a Core Factor in GRN Inference

Data representation plays a central and often underappreciated role in GRN inference. While a large body of existing studies has focused on designing increasingly sophisticated inference algorithms, comparatively less attention has been paid to how gene expression data are encoded prior to inference. In practice, the quality of the inferred network is strongly influenced by the representation of the input data, as it directly determines the amount of biological information preserved, the robustness to noise, and the computational tractability of the inference process.

Traditional GRN inference methods can be broadly categorized according to the form of data representation they employ. Continuous valued representations retain the full resolution of gene expression measurements and are theoretically capable of capturing subtle regulatory effects. However, they are highly sensitive to noise, require large numbers of samples, and often lead to prohibitively high computational complexity, especially for large-scale networks. On the other hand, Boolean representations simplify gene expression into binary states, enabling fast inference and facilitating the modeling of gene dynamics. Despite these advantages, Boolean discretization inevitably leads to information loss and may oversimplify complex regulatory behaviors, particularly when genes exhibit intermediate or graded expression levels.

The findings of this thesis demonstrate that neither extreme fully continuous nor strictly Boolean representations is optimal for GRN inference. Instead, an intermediate representation that balances information preservation and computational efficiency is more suitable. In Chapter 2, we introduced a multiple level discretization strategy in which genes are discretized into either two or three

levels depending on the distribution of their expression values. This adaptive discretization avoids enforcing a uniform representation across all genes and acknowledges the inherent heterogeneity of gene expression dynamics. As a result, biologically meaningful variations are preserved for genes that require finer resolution, while computational efficiency is maintained for genes that can be adequately represented using fewer states.

Furthermore, Chapter 3 extends the notion of data representation beyond discretization by proposing a representation learning framework that integrates gene expression values with structural information. This approach reframes GRN inference as a classification problem, where the success of inference is heavily dependent on how effectively the underlying biological relationships are encoded into feature representations. The improved performance observed in this framework highlights that appropriate representations can enable the use of powerful machine learning models, even when the original biological data are noisy or limited.

Chapter 4 further reinforces the importance of representation by addressing data sparsity through synthetic data generation. By augmenting time-series data using an autoencoder based model, the effective representation of gene dynamics is enriched in the temporal dimension. This not only improves inference accuracy but also stabilizes the learning process when the number of observed time points is small. Importantly, the success of data augmentation depends on the quality of the learned latent representation, again underscoring representation as a key factor in GRN inference.

Taken together, the results presented in this thesis suggest that data representation should be considered a foundational component of GRN inference, rather than a mere preprocessing step. Adaptive discretization, structured feature encoding, and learned latent representations collectively contribute to more accurate, scalable, and biologically meaningful network inference. Future GRN inference studies are therefore likely to benefit from placing greater emphasis on representation design alongside algorithmic development.

5.2.2 Structural Accuracy vs Dynamic Accuracy

Structural accuracy and dynamic accuracy capture two fundamentally different aspects of gene regulatory network inference. Structural accuracy evaluates how well the inferred network topology matches the underlying goldstandard structure, whereas dynamic accuracy assesses the ability of the inferred network to reproduce observed gene expression trajectories. In complex biological systems, these two objectives are not necessarily aligned. Distinct network structures can generate identical or highly similar dynamic behaviors, leading to a multimodal inference landscape.

Our experimental results consistently demonstrate that dynamic accuracy remains high even when structural accuracy degrades, particularly as network size and the number of regulatory inputs increase. This observation highlights the intrinsic ambiguity of GRN inference from timeseries data and suggests that functional validity may be preserved despite structural uncertainty. Consequently, evaluating network inference methods solely based on structural metrics may

be insufficient. Incorporating dynamic accuracy provides a more comprehensive assessment of predictive performance and biological relevance, especially for largescale and densely connected networks.

5.2.3 Scalability and Network Complexity

Scalability remains a critical challenge in gene regulatory network inference, particularly as network size and regulatory complexity increase. As the number of genes grows, the combinatorial search space for potential regulatory interactions expands rapidly, resulting in increased computational cost and inference difficulty. Moreover, higher network density, reflected by a larger number of incoming regulatory links, further exacerbates this challenge by introducing additional ambiguity in identifying true regulatory relationships.

Our results indicate that MIDNI maintains robust dynamic inference performance even in largescale and densely connected networks, despite a gradual decline in structural accuracy. This behavior suggests that the proposed multi-level discretization and information theoretic feature selection strategy effectively mitigates the impact of network complexity. While increased computational cost is unavoidable in large networks, the observed trade-off between scalability and inference accuracy remains favorable, supporting the applicability of MIDNI to realistic biological systems where both network size and regulatory complexity are substantial.

5.3 Limitations of the Proposed Approaches

5.3.1 Multimodality of the Network Inference Problem

A fundamental limitation of gene regulatory network inference lies in the multimodality of the solution space, where multiple distinct network structures are capable of producing similar or even identical dynamic behaviors. This characteristic becomes particularly prominent in large and densely connected networks, in which the number of plausible regulatory configurations increases combinatorially. As a result, optimizing for dynamic accuracy alone does not guarantee the recovery of the true underlying network structure. In this study, MIDNI frequently identified networks that closely reproduced the observed gene expression dynamics while differing from the gold standard topology. This phenomenon reflects an inherent ambiguity in time series based inference rather than a deficiency of the proposed method. Consequently, structural inference accuracy is intrinsically constrained by the limited observability of gene expression data. Addressing this limitation likely requires the integration of additional biological constraints, prior knowledge, or experimental perturbation data to reduce the space of equivalent solutions.

5.3.2 Limitations of Mutual Information-Based Selection

Another important limitation of the proposed framework arises from the use of mutual information as the primary criterion for selecting candidate regulatory

genes. Mutual information is effective in capturing statistical dependencies between gene expression profiles, including nonlinear relationships, but it does not explicitly distinguish between direct and indirect interactions. As a consequence, genes that influence a target gene through intermediate regulators may exhibit high mutual information values and be incorrectly selected as direct regulators. This issue becomes more pronounced in dense networks, where indirect dependencies are common and difficult to disentangle using pairwise or approximate multivariate measures alone. In addition, mutual information does not encode causal directionality, which further complicates accurate regulator identification. While the SWAP subroutine partially alleviates this problem by refining candidate sets based on dynamic consistency, the reliance on mutual information inherently limits structural precision. Incorporating causal inference mechanisms or biological prior knowledge may help mitigate this limitation in future work.

5.3.3 Limitations in Direction and Sign Determination

A further limitation of the proposed approaches concerns the determination of regulatory direction and interaction sign within inferred gene regulatory networks. While the models effectively identify the presence of regulatory relationships, accurately inferring whether a regulation is activating or inhibiting remains challenging, particularly under noisy or sparsely sampled time series conditions. Directionality is often inferred indirectly from temporal ordering or model predictions, yet such cues can be ambiguous when gene expression dynamics exhibit delays, feedback loops, or synchronized behavior. Moreover, classification based frameworks tend to prioritize structural prediction accuracy, which may lead to correct edge detection but incorrect assignment of regulatory sign. This limitation affects the biological interpretability of the inferred networks, since activation and repression convey fundamentally different functional meanings. Although increasing temporal resolution and integrating dynamic modeling can partially improve sign inference, the inherent ambiguity in expression based data constrains reliable direction and sign determination without additional biological constraints or intervention data.

5.3.4 Limitations of Synthetic Data Generation

Despite the effectiveness of synthetic data generation in alleviating data scarcity, this strategy introduces several inherent limitations. The quality of generated samples strongly depends on the amount and diversity of the available original data, and when the initial training set is limited, the learned generative model may fail to capture meaningful temporal patterns, leading to degraded inference performance. In addition, the number of synthesized time points must be carefully controlled, since the generated data at one stage are often used as input for subsequent generation steps, which results in the accumulation of modeling errors over time. As the generation process continues, these errors can propagate and amplify, causing the synthetic data to gradually deviate from realistic gene expression dynamics. Furthermore, synthetic samples may introduce bias by over reinforcing dominant patterns while suppressing rare but biologically

important behaviors. Consequently, excessive reliance on synthetic data can compromise both the accuracy and biological validity of the inferred networks.

5.4 Future Research Directions

5.4.1 Beyond Greedy Search: Advanced Optimization Strategies

The inference framework proposed in this thesis relies on greedy search strategies in the MIFS and SWAP subroutines, which provide computational efficiency and stable performance but are inherently limited when exploring complex and high dimensional solution spaces. Greedy algorithms tend to converge to locally optimal solutions, particularly in multimodal inference problems where multiple regulatory configurations can produce similar dynamic behaviors. To overcome this limitation, future research should investigate advanced optimization strategies that allow broader exploration of the search space while preserving computational feasibility. Potential directions include metaheuristic approaches such as genetic algorithms, simulated annealing, and particle swarm optimization, which have demonstrated effectiveness in navigating multimodal landscapes. In addition, hybrid optimization frameworks that combine greedy initialization with global search refinement may offer a favorable balance between efficiency and accuracy. Incorporating such strategies is expected to improve both structural inference and robustness in large scale gene regulatory networks.

5.4.2 Causal and Mechanistic Inference

Most data driven gene regulatory network inference methods, including the approaches proposed in this thesis, primarily focus on statistical dependencies rather than true causal and mechanistic relationships among genes. Although inferred interactions may exhibit high structural or dynamical accuracy, they do not necessarily reflect direct regulatory effects or underlying biological mechanisms. This limitation becomes more pronounced when genes are indirectly connected through hidden regulators or unobserved confounding factors. Future research should aim to integrate causal inference frameworks that explicitly model intervention effects and temporal precedence, allowing a clearer distinction between correlation and causation. In addition, incorporating mechanistic constraints derived from transcription factor binding, signaling pathways, or kinetic models may enhance biological interpretability. A hybrid framework that combines machine learning based inference with causal reasoning and mechanistic knowledge could lead to more reliable and biologically meaningful gene regulatory networks.

5.4.3 Integration of Prior Biological Knowledge

While the proposed methods rely on data driven learning from gene expression profiles, incorporating prior biological knowledge represents an important direction for improving the reliability and interpretability of inferred gene regulatory

networks. Purely data driven approaches may suffer from false positive interactions, especially when expression data are noisy, sparse, or limited in temporal resolution. Integrating information from biological databases such as known transcription factor target relationships, protein DNA binding data, or curated signaling pathways can help constrain the search space and guide the inference process toward biologically plausible solutions. Prior knowledge can be incorporated in various forms, including regularization terms, probabilistic priors, or post inference filtering strategies. Such integration not only has the potential to improve structural accuracy but also enhances biological credibility, enabling inferred networks to better reflect true regulatory mechanisms rather than statistical artifacts.

5.4.4 Extension to Other Data Types

All three methods proposed in this thesis are developed and evaluated based on time series gene expression data, which provide valuable temporal information for capturing regulatory dynamics. However, another widely available and biologically relevant data type is steady state gene expression data, which reflect equilibrium conditions under different perturbations or environmental settings. Steady state data are often easier to obtain and can complement time series observations, especially when the number of observed time points is limited. A promising future direction is to design an integrated data representation framework that jointly exploits both time series and steady state expression data. By combining dynamic temporal patterns with steady state dependencies, such a unified representation has the potential to enhance network inference robustness and accuracy, particularly in scenarios where either data type alone is insufficient.

5.5 Concluding Remarks

This thesis investigated the problem of gene regulatory network inference from gene expression data, with a particular focus on improving inference performance through effective data representation and learning strategies. Across the three proposed frameworks, we demonstrated that how gene expression data are represented plays a critical role in determining both structural and dynamical inference accuracy. By moving beyond conventional Boolean representations and raw real valued data, this work introduced flexible multi level discretization, representation learning that integrates expression and structural information, and synthetic data generation to address data scarcity in time series settings.

Extensive experiments on benchmark datasets showed that the proposed methods consistently outperformed existing approaches, especially for large scale and complex networks. These results highlight the importance of adapting inference strategies to data characteristics rather than relying on fixed modeling assumptions. Although several limitations remain, including optimization challenges and dependency on statistical measures, this thesis provides a unified perspective on network inference that bridges data representation, machine learning, and biological relevance.

Overall, this work contributes methodological insights and practical frameworks that can serve as a foundation for future advances in robust and scalable gene regulatory network inference.

Appendix A

Supplementary Materials for Chapter 3

A.1 Algorithm for Synthetic Dataset Generation Based on the Barabási–Albert Model

Algorithm S1

Algorithm 1 RandomUpdateFunction

Require: Graph $G(V, A)$
1: **for each** $v \in V$ **do**
2: let $u_1, u_2, \ldots, u_k$ genes regulating v in G
3: let $l_1, l_2, \ldots, l_k, l$ the discretization level of $u_1, u_2, \ldots, u_k$ and v
4: **for each** $x \in \{0, \ldots, l_1 - 1\} \times \{0, \ldots, l_2 - 1\} \times \cdots \times \{0, \ldots, l_k - 1\}$ **do**
5: $F_v(x) \leftarrow \text{randint}(0, l - 1)$ ▷ a random integer between 0 and $l - 1$
6: **end for**
7: **end for**
8: **return** $\{F_v | v \in V\}$

Algorithm S2

A.2 Detailed performance results on the *E. coli* dataset

This section provides the complete quantitative results underlying the performance analysis reported in the subsection "Performance on *E. coli* datasets" (see in the main text). While the main text focuses on aggregated trends and comparative observations, the tables presented here report detailed AUPR scores for individual network samples and time series across different network sizes. Specifically, Tables A.1, A.2, A.3, and A.4 summarize the performance of all evaluated methods on networks of size 10, 50, and 100, respectively. These results are included to support reproducibility, facilitate fine-grained comparison among methods, and provide additional insight into performance variability across samples and experimental settings.

Algorithm 2 RandomGeneExpression

Require: Graph $G(V, A)$, update functions F_v, observed time step K

for each $v \in V$ **do**
  let the discretization level of v
  $v(0) \leftarrow \text{randint}(0, l-1)$ ▷ a random integer between 0 and $l-1$
end for
for $t = 1$ to K **do**
  for each $v \in V$ **do**
    let v: $u_1, u_2, \ldots, u_k$ genes regulating v in G
    $x = (u_1(t-1), u_2(t-1), \ldots, u_k(t-1))$
    $v(t+1) \leftarrow F_v(x)$
  end for
end for
return $\{v(t) | v \in V, t \in \{0, \ldots, K\}\}$

TABLE A.1: AUPR performance of different inference methods on the *E. coli* dataset with network size 10 (Samples 1–5).

Model	Network size 10 (Samples 1–5)									
	Sample 1		Sample 2		Sample 3		Sample 4		Sample 5	
	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2
LR	0.3077	0.3852	0.5803	0.6818	0.8626	0.9311	0.9292	0.9760	0.5277	0.5162
DT	0.3000	0.3179	0.3558	0.3808	0.6778	0.4760	0.1714	0.2722	0.4174	0.3253
RF	**0.6873**	0.2810	0.5460	**0.7145**	0.9111	0.8111	0.9111	0.9167	0.6778	0.5242
KNN	0.5315	0.2683	0.3705	0.6684	0.5639	0.4861	0.8017	0.6222	**0.7485**	0.4899
XGBoost	0.6236	0.6370	0.5338	0.6964	0.9229	**0.9826**	0.9271	**1.0000**	0.4439	0.4576
LGBM	0.6603	**0.6395**	0.5915	0.6998	0.9181	0.9602	0.9404	0.9905	0.4429	0.5047
MLP	0.4376	0.3069	**0.6487**	0.6322	**0.9322**	0.7100	0.9247	0.9826	0.5258	**0.5920**
GNB	0.5099	0.5099	0.6327	0.6327	0.6356	0.6107	**0.9556**	**0.9556**	0.5148	0.5503
Baseline	0.1663	0.1746	0.1572	0.1594	0.1398	0.1360	0.1368	0.1356	0.2696	0.2639
Jump3	0.1946	0.1946	0.1398	0.1398	0.1097	0.1097	0.1041	0.1041	0.2373	0.2373
Inferelator	0.1439	0.1213	0.1679	0.1905	0.0970	0.1161	0.1153	0.1815	0.2518	0.2945
CLR-lag	0.1653	0.1829	0.2138	0.2177	0.1145	0.1702	0.1136	0.1993	0.2820	0.2561
TSNI	0.1429	0.1537	0.1732	0.1434	0.1982	0.2621	0.3033	0.1701	0.3154	0.2206
G1DBN	0.1485	0.1803	0.1590	0.1443	0.1736	0.1115	0.2633	0.1287	0.2523	0.3205

TABLE A.2: AUPR performance of different inference methods on the *E. coli* dataset with network size 10 (Samples 6–10 and average).

Model	Network size 10 (Samples 6–10)										Average
	Sample 6		Sample 7		Sample 8		Sample 9		Sample 10		
	S1	S2	S1	S2	S1	S2	S1	S2	S1	S2	
LR	**0.6400**	**0.7022**	0.6465	0.2051	0.7677	0.7876	0.1254	0.1089	0.2276	0.2083	0.5559
DT	0.2808	0.3344	0.1381	**0.6106**	0.3804	0.5342	0.1263	0.1208	0.2428	0.2059	0.3334
RF	0.4399	0.5709	0.6183	0.1548	0.7376	0.7376	0.4000	0.0500	0.1056	0.1056	0.5450
KNN	0.4915	0.3535	0.4240	0.1464	0.7614	0.6126	0.2611	0.2000	0.1680	0.1819	0.4576
XGBoost	0.3621	0.5192	0.3097	0.2175	0.9007	0.8931	0.1266	0.1104	0.1735	0.1800	0.5509
LGBM	0.4043	0.5452	**0.9024**	0.2299	**0.9106**	0.8946	0.1278	0.1170	0.1632	0.1873	**0.5915**
MLP	0.5673	0.5815	0.3363	0.3082	0.9075	**0.9180**	0.1129	0.1128	0.1645	0.1847	0.5443
GNB	0.4207	0.4551	0.6061	0.6094	0.6309	0.6550	**0.4017**	**0.2877**	**0.3713**	**0.3773**	0.5661
Baseline	0.2107	0.2017	0.1019	0.0988	0.2187	0.2160	0.1234	0.1164	0.2266	0.2368	0.1745
Jump3	0.3195	0.3195	0.0907	0.0907	0.2077	0.2077	0.0924	0.0924	0.2430	0.2430	0.1739
Inferelator	0.1793	0.2938	0.0389	0.0624	0.2762	0.1179	0.1323	0.0936	0.2135	0.2925	0.1690
CLR-lag	0.2192	0.3492	0.0829	0.1050	0.2123	0.2171	0.1212	0.1261	0.2857	0.2662	0.1950
TSNI	0.3722	0.2487	0.1820	0.1231	0.2666	0.2784	0.1306	0.1296	0.2241	0.2351	0.2137
G1DBN	0.2808	0.2084	0.1033	0.0825	0.2297	0.2759	0.1000	0.1000	0.2331	0.1983	0.1847

TABLE A.3: AUPR performance of different inference methods on the *E. coli* dataset with network size 50. Models are trained and evaluated on networks of the same size. Results are reported for four independent network samples, each evaluated on two test time series. The final column reports the average AUPR across all samples. The best performing method in each data is highlighted in bold.

Model	Network size 50 (Samples and time series)								Average
	Sample 1		Sample 2		Sample 3		Sample 4		
	S1	S2	S1	S2	S1	S2	S1	S2	
LR	0.1619	0.0703	0.0931	0.1441	0.0732	0.0955	0.0761	0.0663	0.0976
DT	0.1043	0.2552	0.1338	0.1849	0.1856	0.3494	0.1911	0.1508	0.1944
RF	0.0593	0.0182	0.1664	**0.3088**	0.0692	0.0188	0.1762	0.1433	0.1200
KNN	0.1859	0.0540	0.0897	0.0905	0.0432	0.0419	0.1463	0.1406	0.0990
XGBoost	0.1498	0.0757	0.1004	0.1884	**0.2365**	0.0923	0.1192	0.1148	0.1346
LGBM	**0.4383**	0.0765	0.1100	0.1988	0.2242	0.0936	0.1345	0.1583	0.1793
MLP	0.2958	0.0785	0.1150	0.1764	0.0922	0.0637	0.1216	0.1295	0.1341
GNB	0.4207	**0.4790**	**0.2996**	0.2929	**0.4240**	**0.3910**	**0.2697**	**0.2657**	**0.3553**
Baseline	0.0378	0.0373	0.0445	0.0434	0.0398	0.0392	0.0495	0.0502	0.0427
Jump3	0.0451	0.0451	0.0475	0.0475	0.0589	0.0589	0.0492	0.0492	0.0502
Inferelator	0.0406	0.0352	0.0477	0.0517	0.0293	0.0387	0.0400	0.0364	0.0400
CLR-lag	0.0409	0.0350	0.0472	0.0401	0.0364	0.0508	0.0546	0.0534	0.0448
TSNI	0.0482	0.0334	0.0498	0.0451	0.0442	0.0590	0.0506	0.0471	0.0472
G1DBN	0.0400	0.0373	0.0439	0.0450	0.0387	0.0396	0.0481	0.0490	0.0427

TABLE A.4: AUPR performance of different inference methods on the *E. coli* dataset with network size 100. Models are trained and evaluated on networks of the same size. Results are reported for four independent network samples, each evaluated on two test time series. The final column reports the average AUPR across all samples. The best performing method in each data is highlighted in bold.

Model	Network size 100 (Samples and time series)								Average
	Sample 1		Sample 2		Sample 3		Sample 4		
	S1	S2	S1	S2	S1	S2	S1	S2	
LR	0.0332	0.0307	0.0391	0.0374	0.0521	0.0563	0.0336	0.0374	0.0400
DT	0.3146	0.2220	0.1845	0.2078	0.1674	0.1553	0.1695	0.2176	0.2048
RF	0.1745	0.0461	0.1309	0.1546	0.1171	0.1357	0.1024	0.1019	0.1204
KNN	0.0124	0.0124	0.0858	0.0124	0.0124	0.1394	0.1163	0.0123	0.0504
XGBoost	0.0595	0.0312	0.0600	0.0446	0.0469	0.0645	0.0473	0.0515	0.0507
LGBM	0.0611	0.0325	0.0493	0.0462	0.0582	0.0576	0.0428	0.0458	0.0492
MLP	0.0454	0.0317	0.0430	0.0422	0.0490	0.0522	0.0489	0.0628	0.0469
GNB	**0.3579**	**0.3511**	**0.3086**	**0.3018**	**0.3817**	**0.3897**	**0.2571**	**0.2424**	**0.3238**
Baseline	0.0254	0.0251	0.0254	0.0251	0.0252	0.0249	0.0250	0.0252	0.0252
Jump3	0.0390	0.0390	0.0281	0.0281	0.0280	0.0280	0.0282	0.0282	0.0308
Inferelator	0.0280	0.0239	0.0281	0.0255	0.0212	0.0364	0.0244	0.0275	0.0269
CLR-lag	0.0235	0.0251	0.0269	0.0270	0.0260	0.0254	0.0238	0.0233	0.0251
TSNI	0.0279	0.0250	0.0266	0.0290	0.0330	0.0259	0.0285	0.0264	0.0278
G1DBN	0.0255	0.0249	0.0261	0.0262	0.0277	0.0324	0.0246	0.0255	0.0266

A.3 Effect of Training Data Size and Test Network Scale on GNB Performance

To support the analysis presented in the main manuscript, this section reports detailed experimental results examining the effect of training data size and test network scale on the predictive performance of the GNB model. Detailed quantitative results for each experimental setting are presented in Tables A.5, A.6, and A.7.

TABLE A.5: Detailed AUPR results of the GNB model on size-10 networks with testing size 10. For each training set size, the highest AUPR value across time series realizations is highlighted in bold.

Sample	TS	5k	10k	20k
1	S1	0.5159	0.5159	0.5099
	S2	0.5159	0.5159	0.5099
2	S1	0.6383	0.6383	0.6327
	S2	0.6383	0.6383	0.6327
3	S1	0.7203	0.7770	0.6356
	S2	0.6601	0.6806	0.6107
4	S1	**0.9556**	**0.9556**	**0.9556**
	S2	**0.9556**	**0.9556**	**0.9556**
5	S1	0.5080	0.5376	0.5148
	S2	0.5542	0.5627	0.5503
6	S1	0.3925	0.3910	0.4207
	S2	0.4470	0.4456	0.4551
7	S1	0.6061	0.6061	0.6061
	S2	0.6094	0.6094	0.6094
8	S1	0.6398	0.6267	0.6309
	S2	0.6550	0.6497	0.6550
9	S1	0.2828	0.2817	0.4017
	S2	0.2857	0.2817	0.2877
10	S1	0.4470	0.4470	0.3713
	S2	0.4271	0.4530	0.3773
Average		0.5727	0.5785	0.5661

TABLE A.6: Detailed AUPR results of the GNB model with testing size 50. For each training set size, the highest AUPR value across time series realizations is highlighted in bold.

Sample	TS	5k	10k	20k
1	S1	0.3820	0.4044	0.4207
	S2	**0.4756**	**0.4806**	**0.4790**
2	S1	0.3009	0.3016	0.2996
	S2	0.2877	0.2944	0.2929
3	S1	0.4260	0.4252	0.4240
	S2	0.3811	0.3809	0.3910
4	S1	0.2796	0.2703	0.2697
	S2	0.2571	0.2568	0.2657
Average		0.3487	0.3518	0.3553

TABLE A.7: Detailed AUPR results of the GNB model with testing size 100. For each training set size, the highest AUPR value across time series realizations is highlighted in bold.

Sample	TS	5k	10k	20k
1	S1	0.3612	0.3634	0.3579
	S2	0.3607	0.3608	0.3511
2	S1	0.2975	0.3040	0.3086
	S2	0.3014	0.3054	0.3018
3	S1	0.3656	0.3758	0.3817
	S2	**0.3799**	**0.3817**	**0.3897**
4	S1	0.2585	0.2586	0.2571
	S2	0.2540	0.2499	0.2424
Average		0.3224	0.3249	0.3238

Appendix B

Supplementary Materials for Chapter 4

Detailed results of the performance on multi-level and Boolean discretization datasets

K	10	20	30	40	50	60	70	80	90
Ground truth networks, size = 50									
Precision	0.298 ±0.054	0.468 ±0.069	0.569 ±0.068	0.637 ±0.090	0.686 ±0.085	0.728 ±0.084	0.744 ±0.077	0.768 ±0.084	0.783 ±0.087
Recall	0.224 ±0.045	0.464 ±0.069	0.618 ±0.070	0.706 ±0.077	0.759 ±0.065	0.797 ±0.069	0.808 ±0.063	0.829 ±0.062	0.845 ±0.072
Structural	0.940 ±0.006	0.950 ±0.008	0.960 ±0.008	0.967 ±0.010	0.972 ±0.009	0.977 ±0.008	0.978 ±0.007	0.980 ±0.008	0.982 ±0.008
Dynamics	0.997 ±0.005	0.999 ±0.001	0.997 ±0.005	0.995 ±0.009	0.995 ±0.008	0.991 ±0.011	0.989 ±0.009	0.987 ±0.009	0.988 ±0.007
Synthetic networks, size = 50									
Precision	0.397 ±0.042	0.521 ±0.086	0.586 ±0.072	0.650 ±0.089	0.689 ±0.093	0.725 ±0.071	0.772 ±0.093	0.779 ±0.076	0.788 ±0.080
Recall	0.418 ±0.023	0.532 ±0.086	0.638 ±0.076	0.719 ±0.116	0.760 ±0.102	0.788 ±0.114	0.818 ±0.103	0.831 ±0.069	0.846 ±0.075
Structural	0.957 ±0.011	0.954 ±0.015	0.965 ±0.014	0.974 ±0.017	0.972 ±0.016	0.976 ±0.012	0.980 ±0.014	0.981 ±0.010	0.981 ±0.010
Dynamics	0.999 ±0.001	0.999 ±0.001	0.997 ±0.005	0.995 ±0.007	0.990 ±0.011	0.985 ±0.014	0.984 ±0.008	0.979 ±0.010	0.985 ±0.009
Ground truth networks, size = 100									
Precision	0.215 ±0.029	0.416 ±0.054	0.502 ±0.064	0.587 ±0.054	0.627 ±0.060	0.674 ±0.046	0.701 ±0.053	0.724 ±0.077	0.729 ±0.084
Recall	0.161 ±0.024	0.407 ±0.052	0.557 ±0.063	0.671 ±0.050	0.729 ±0.055	0.774 ±0.041	0.797 ±0.043	0.814 ±0.059	0.814 ±0.064
Structural	0.967 ±0.002	0.973 ±0.003	0.977 ±0.004	0.981 ±0.003	0.984 ±0.004	0.986 ±0.003	0.987 ±0.003	0.988 ±0.004	0.989 ±0.004
Dynamics	0.999 ±0.001	1.000 ±0.001	0.999 ±0.000	0.998 ±0.002	0.993 ±0.008	0.988 ±0.011	0.983 ±0.013	0.982 ±0.012	0.979 ±0.014
Synthetic networks, size = 100									
Precision	0.286 ±0.033	0.426 ±0.038	0.565 ±0.085	0.599 ±0.090	0.641 ±0.048	0.685 ±0.092	0.707 ±0.065	0.731 ±0.075	0.741 ±0.063
Recall	0.395 ±0.022	0.476 ±0.048	0.571 ±0.079	0.699 ±0.105	0.729 ±0.076	0.798 ±0.101	0.817 ±0.106	0.819 ±0.100	0.823 ±0.068
Structural	0.981 ±0.003	0.976 ±0.004	0.979 ±0.008	0.984 ±0.009	0.982 ±0.004	0.988 ±0.008	0.985 ±0.006	0.987 ±0.006	0.983 ±0.005
Dynamics	1.000 ±0.001	0.998 ±0.004	0.998 ±0.002	0.991 ±0.010	0.980 ±0.010	0.977 ±0.017	0.969 ±0.018	0.970 ±0.013	0.972 ±0.016

TABLE B.1: Performance metrics (mean ± standard deviation) on multi-level discretization datasets across varying numbers of observed time steps K.

K	10	20	30	40	50	60	70	80	90
Ground truth networks, size = 50									
Precision	0.377 ±0.089	0.449 ±0.043	0.524 ±0.071	0.579 ±0.067	0.606 ±0.083	0.615 ±0.093	0.627 ±0.101	0.644 ±0.126	0.637 ±0.126
Recall	0.249 ±0.058	0.441 ±0.044	0.540 ±0.070	0.591 ±0.071	0.617 ±0.078	0.617 ±0.090	0.626 ±0.090	0.640 ±0.114	0.638 ±0.113
Structural	0.947 ±0.008	0.950 ±0.005	0.956 ±0.007	0.962 ±0.007	0.964 ±0.008	0.965 ±0.010	0.966 ±0.010	0.967 ±0.013	0.966 ±0.013
Dynamics	0.997 ±0.003	0.998 ±0.003	0.991 ±0.003	0.990 ±0.004	0.989 ±0.005	0.991 ±0.005	0.992 ±0.003	0.994 ±0.003	0.994 ±0.003
Synthetic networks, size = 50									
Precision	0.382 ±0.057	0.467 ±0.054	0.538 ±0.103	0.584 ±0.127	0.607 ±0.133	0.630 ±0.119	0.632 ±0.123	0.643 ±0.132	0.643 ±0.120
Recall	0.337 ±0.059	0.475 ±0.053	0.573 ±0.074	0.621 ±0.080	0.645 ±0.080	0.645 ±0.095	0.647 ±0.101	0.648 ±0.104	0.645 ±0.110
Structural	0.955 ±0.009	0.957 ±0.010	0.961 ±0.015	0.966 ±0.020	0.967 ±0.020	0.966 ±0.017	0.969 ±0.016	0.968 ±0.019	0.968 ±0.016
Dynamics	1.000 ±0.000	0.999 ±0.000	0.995 ±0.002	0.990 ±0.006	0.989 ±0.007	0.990 ±0.006	0.991 ±0.005	0.989 ±0.008	0.990 ±0.006
Ground truth networks, size = 100									
Precision	0.314 ±0.054	0.409 ±0.030	0.471 ±0.047	0.497 ±0.041	0.561 ±0.050	0.588 ±0.056	0.613 ±0.048	0.626 ±0.052	0.643 ±0.058
Recall	0.187 ±0.029	0.377 ±0.022	0.490 ±0.041	0.527 ±0.029	0.585 ±0.051	0.607 ±0.052	0.625 ±0.056	0.635 ±0.061	0.648 ±0.070
Structural	0.972 ±0.002	0.973 ±0.002	0.976 ±0.003	0.977 ±0.003	0.980 ±0.003	0.981 ±0.003	0.982 ±0.003	0.983 ±0.003	0.984 ±0.003
Dynamics	0.996 ±0.003	0.999 ±0.001	0.995 ±0.002	0.989 ±0.004	0.988 ±0.005	0.988 ±0.006	0.989 ±0.005	0.990 ±0.005	0.991 ±0.004
Synthetic networks, size = 100									
Precision	0.381 ±0.039	0.429 ±0.041	0.485 ±0.055	0.535 ±0.072	0.593 ±0.080	0.611 ±0.085	0.638 ±0.089	0.642 ±0.066	0.647 ±0.101
Recall	0.228 ±0.022	0.377 ±0.034	0.495 ±0.039	0.558 ±0.049	0.621 ±0.027	0.636 ±0.091	0.647 ±0.072	0.649 ±0.062	0.649 ±0.063
Structural	0.980 ±0.005	0.981 ±0.005	0.983 ±0.006	0.983 ±0.008	0.984 ±0.007	0.985 ±0.007	0.985 ±0.007	0.985 ±0.005	0.985 ±0.007
Dynamics	0.999 ±0.003	0.998 ±0.002	0.993 ±0.004	0.988 ±0.005	0.986 ±0.006	0.980 ±0.006	0.982 ±0.007	0.983 ±0.007	0.984 ±0.009

TABLE B.2: Performance metrics (mean ± standard deviation) on Boolean datasets across varying numbers of observed time steps K.

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